MILITARY SPECIFICATION
MANUALS, TECHNICAL: ORGANIZATIONAL (FLIGHT LINE) MAINTENANCE FOR REMOTELY PILOTED VEHICLES (RPV)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 The specification covers the requirements for preparation of technical manuals for organizational (flight line) maintenance instructions for remotely piloted vehicles (RPV).

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on the date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-M-38784	Manuals, Technical; General Requirements for preparation of

STANDARDS

Military

MIL-STD-12	Abbreviations for Use on Drawings and in Technical Type Publications
MIL-STD-863	Wiring Data, Preparation of
ANSI Y.32.2-1975	Graphic Symbols for Electrical and Electronic Diagrams
MIL-STD-122	Mechanical Symbols for Aeronautical, Aerospacecraft, and Spacecraft Use
ANSI Y.32.9-1972	Electrical Wiring and Layout Diagrams used in Architecture and Building Construction, Graphic Symbols For
ANSI Y.32.16.1975	Reference Designations for Electrical and Electronic Parts and Equipments

AIR FORCE PUBLICATIONS

Technical Orders

T.O. 00-5-1	AF Technical Order System
T.O. 00-5-15	AF Time Compliance Technical Order System

3. REQUIREMENTS

3.1 General Requirements. In event of conflict between governing directives, the contract, this specification, and MIL-M-38784 shall take precedence over other directives in the order stated.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: AFLC/LOLMP, WPAFB OH, 45433 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

TMSS

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3.1. Number and Types of Manuals. The following manuals shall be provided, the scope, content and arrangement of which are covered in subsequent paragraphs giving the requirements for each type:

(a) RPV General Manual.

(b) Ground Handling, Servicing, and Airframe Maintenance Manual.

(c) Uncrating, Crating, Assembly, and Disassembly Manual.

(d) Power Plant Manual.

(e) Environmental, Fuel and Oil System Manual.

(f) Recovery System Manual.

(g) Electrical Power Supply and Distribution Manual.

(h) Electronics and Guidance Systems Manual.

(i) Flight Control Systems Manual.

(j) Utility Systems Manual.

(k) Decontamination Manual.

(l) Upload, Pre/Postflight and Download Manual.

(m) Wiring Diagram Manual.

(n) Operational Systems Test Manual.

(o) Payload and Reconnaissance System Manual.

3.1.2 Additional Manuals. Additional manuals may be provided at the option of the procurement activity. These manuals may be required to support organizational maintenance of peculiar payload systems or additional design features.

3.1.3 Coverage in a Single Manual. If the RPV is relatively uncomplicated, the organizational (flight line) maintenance instructions may be covered in a single manual if approved by the procuring activity. The technical content of such a manual shall be the same as if covered in the several manuals specified in paragraph 3.1.1. The sections of such a manual shall constitute what the manuals specified in paragraph 3.1.1 would constitute; the main paragraphs of such a manual shall constitute what the sections of manuals specified in paragraph 3.1.1 would constitute.

3.1.4 General Manner of Preparation. The general manner of preparation of each of the manuals specified herein shall be in accordance with Specification MIL-M-38784 except as otherwise specified herein. Each of these manuals shall be considered an independent manual for purposes of format, for example, each shall have a title page, "A" page, foreword, table of contents, separate sections, etc.

3.1.5 Text Content. Collectively, the manuals shall contain all essential information required by personnel for performing organizational (flight line) maintenance of the RPV. Each manual shall contain all instructions for the work involved except the RPV General Manual recognizing that some of the data will be duplicated or partially duplicated in other manuals specified herein.

3.1.6 Style of Text. The style of the text shall be as specified in subsequent paragraphs covering the various types of manuals specified herein.

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3.1.7 Format. Each manual shall be complete, for example, each shall have a foreword, table of contents, have its pages numbered independently of other manuals, have its own date, and have other format requirements for a manual as specified in Specification MIL-M-38784.

3.1.7.1 Each section of each manual shall begin on a right-hand page.

3.1.7.2 The pages of each manual shall be numbered per MIL-M-38784.

3.1.7.3 Tables, charts, and other illustrations shall be numbered separately for each manual. Numbering will be per MIL-M-38784.

3.1.7.4 General requirements of MIL-M-38784 related to supersedure notice, changes, revisions, supplements, and numbering shall apply.

3.1.8 Front Matter. The front matter of each manual shall consist of the following data arranged in the order listed:

(a) Title page prepared per MIL-M-38784.

(b) List of effective pages prepared per MIL-M-38784.

(c) A configuration page prepared per figure 1.

(d) A foreword containing the information in paragraph 3.1.8.1.

(e) A station locator diagram prepared per figure 2.

(f) A safety summary covering hazards involved in the manual. The safety summary shall be prepared per MIL-M-38784.

(g) A single-column table of contents prepared per MIL-M-38784.

(h) A single-column list of illustrations prepared per MIL-M-38784.

(i) A single-column list of tables prepared per MIL-M-38784.

3.1.8.1 Foreword. Each manual shall have a foreword. The foreword shall precede the table of contents, and shall include the series and model of the RPV, the established nomenclature of the equipment covered, give the specific use to be made of the manual, make reference to other related manuals specified herein, and give any relevant information that will be of assistance in the use of the manual. If there is a supplement to the manual, the foreword shall identify such a supplement and give the assigned technical order number and use to be made of the supplement. The foreword shall contain a specific list of manuals required by this specification, a brief statement of the purpose and application of the manuals, a list of applicable publications, an explanation of Warnings, Cautions and Notes, explanation of standard and nonstandard abbreviations, and a glossary of terms. A statement shall be included to reference T.O. 00-5-1 for use and application of the technical order and T.O. 00-20- series publications for documentation requirements.

3.1.9 Classified Instructions. Where classified instructions will comprise approximately 10 percent or less of a total manual, and will be concentrated in one portion of the manual, such instructions shall be prepared in the form of a separate manual in order that the basic manual may be unclassified. The format and content of such instructions shall be the same as if the instructions were prepared for inclusion in the basic manual. There shall be an introduction in the classified manual explaining that the classified manual is incomplete without the basic manual and their relationship. The procuring activity may direct this classified information to be placed in the RPV General Manual in lieu of preparing a separate manual. This will require additional

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sections be added to the RPV General Manual.

3.1.10 Security Classification. The security classification of the manuals shall be as set forth in the Program Security Classification Guide with procedures in accordance with Specification MIL-M-38784.

3.1.11 Illustrations. Illustrations shall be provided in accordance with specification MIL-M-38784 and as required herein.

3.1.12 Nomenclature. Nomenclature of items, though in short, descriptive form, shall be definite enough to ensure clear identification. The principle noun in names of items shall be consistent with those on engineering drawings and shall be the same throughout the manuals. Part numhers shall be held to an absolute minimum and shall be used only when no other means of positive identification will suffice.

3.1.13 Model or Type Reference. The following, in addition to requirements for model or type referenced in MIL-M-38784, shall apply: If no other method of identifying changes on applicable equipment is practical, the time compliance technical order number authorizing the change may be included.

3.1.14 Electrical and Electronic Symbols. Drawing graphic symbols for electrical and electronics parts shall be in accordance with ANSI Y.32.2-1970, ANSI Y.32.9-1972, and MIL-STD-863.

3.1.5 Reference Designations. Reference designations for electrical symbols shall be in accordance with Standard USAS Y.32.16-1968.

3.1.16 Color Coding. Reference to color coding of chassis wiring shall be in accordance with Standard MIL-STD-122. Chassis wiring in equipment which does not conform to requirements of Standard MIL-STD-122 shall be explained in the manual.

3.1.17 Abbreviations. The use of abbreviations shall be kept to an absolute minimum. Standard abbreviations used shall be in accordance with Standard MIL-STD-12. Nonstandard abbreviations shall be identified in the forword of applicable manual.

3.1.18 Type of Instructions to be Excluded. Instructions shall not cover maintenance normally performed by intermediate (field) level maintenance personnel, for example:

(a) Complete disassembly and reassembly of any component, subassembly, assembly, accessory, or plug-in unit.

(b) Reconditioning, repair, or replacement instructions which require tools, test equipment not authorized to organizational (flight line) maintenance activities by applicable tables of allowances.

(c) Bench check of removed components.

3.1.19 Configuration Coding. Two or more RPV configurations of the same mission design series (MDS) may be included in any of the organizational maintenance manuals specified herein. Data peculiar to a single configuration RPV shall be coded for ease of identification. Code letters (A B C etc.) shall be assigned to each configuration. The applicable code letter shall appear adjacent to the paragraph procedural step, test response, figure, table, index, part number, warning, caution, or note data. Data applicable to all configurations shall not be coded. The coding assignments, use, and application shall be defined in the foreword of applicable manuals. See figure 15 for sample wording.

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3.1.20 RELATIONSHIP TO OTHER MANUALS

3.1.20.1 Inspection Manuals. The organizational maintenance manuals prescribed by this specification are interrelated to the RPV inspection manuals (-6 series). The organizational maintenance manuals must provide instructions for accomplishing inspection requirements established by the inspection manuals and for replacing components listed in the replacement schedule of the inspection manual.

3.1.20.2 Illustrated Parts Breakdown. The illustrated parts breakdown (-4 series) contains source, maintenance and repair codes (SMR) for each replaceable component comprising the RPV. The organizational maintenance manual must include instructions for removal, installation, and repair of items coded for such actions at the organizational level if such procedures are not obvious.

3.1.20.3 Work Unit Code Manual. The work unit code manual (-06 series) identifies the replaceable components in the system in which they primarily function. Normally the organizational maintenance manuals will include the components/equipments in the manual covering the system in which they are coded in the work unit code manual. If a component functions in more than one system, it will be included in the manual covering the system in which it is coded and be referenced in other systems in which it functions.

3.2 General Guidelines for Preparation of Manuals Specified Herein. The following instructions apply to all manuals specified herein insofar as detailed manual outlines contained in paragraphs 3.3 through 3.15 require the information as part of the manual.

3.2.1 Scope of Information to be Included. Each manual shall provide all essential information required by personnel for performing organizational (flight line) maintenance of the systems and equipment. Each system or equipment shall be treated functionally. Equipments which function in more than one system shall be included in the manual covering the primary function and referenced in manuals covering other systems in which they function (i.e., the starter-generator shall be included in the power plant manual and referenced in the electrical power supply and distribution manual). The maintenance instructions shall be provided so as to enable personnel to connect the test equipment and perform operational checkout which will determine if the system meets its installed minimum performance standards. Instructions shall include descriptions, installation and removal procedures and procedures for inspecting, cleaning, adjusting, lubricating, and servicing the system or equipment. Maintenance requirements for extreme climatic conditions shall be included. Instructions shall cover all aspects of a system or equipment and portions of other systems or equipments utilized as part of the system under consideration. For example, coverage of the flight control system shall include coverage of the control circuitry, mechanical interface, electrical, and other aspects of the system. Reference to the applicable Work Unit Code manual will aid in determining primary functions.

3.2.1.1 Equipment Description. A complete physical description shall be provided for each equipment or system covered in the manual. The description shall include leading particulars, size, weight, mounting features, location in the RPV, and mechanical interface. The physical description shall be written in such a manner, supported by illustrations as required, to enable personnel to identify the actual equipment. A separate functional description shall be provided to facilitate understanding of the system or equipment. This shall describe the complete system or equipment, data flow, and related mechanical and electrical aspects. The functional description will normally be supported by system or equipment block diagram.

3.2.1.2 Identification of Support Equipment and Materials. Most replacement, maintenance, and servicing actions require the use of handling, servicing, or test equipment, special tools and consumable materials. Each section of the manual that covers installation, removal, maintenance or servicing shall

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identify the required equipment and materials in tabular format. All equipment and tools required to perform each maintenance action, test, or adjustment shall be listed at the beginning of the section prescribing such work. Common types of test equipment such as voltmeters, generators, tube testers, etc., shall be listed. Common types of tools such as screwdrivers, pliers, side cutters, soldering irons, drills, etc., shall not be listed. Consumable materiels, such as lubricants, cleaning solvents, test fluids, wiping cloth, etc., shall be listed. If no consumable materials are required the subordinate paragraph covering these items shall be omitted. The primary paragraph, EQUIPMENT REQUIRED, is mandatory and shall be included even when equipment is not required. If no equipment is required a statement to that affect shall be included. Listings shall be as follows:

Equipment Required

Item	Figure and Index No.	Nomenclature	*AN Type Designation	**Alternate	Use and Application

* If none, then the manufacturer's designation
** If none, then enter the word none

Tools Required

Item	Figure and Index No.	Part Number	Nomenclature	Use and Applicatoion

Consumable Materials Required

Item	Part Number	Specification	Nomenclature	Use and Application

3.2.1.2.1 Figure and Index Column. Include the applicable figure and index number that illustrates the equipment or tool. If the item is not illustrated this column will be left blank. If none of the equipment listed is illustrated the entire column may be omitted from the table.

3.2.1.2.2 Part Number Column. Equipment and materials shall be listed in numerical order by part number.

3.2.1.2.3 Nomenclature Column. Nomenclature shall be identical with that used on the engineering drawings or applicable specifications.

3.2.1.2.4 Use and Application Column. If the section covers more than one equipment model, the application to the various models shall be shown in this column. In addition, state the specific purpose of each item. The use and application column may be divided as follows to reflect multiple applications. If this method is elected the application codes will be identified in a legend at the bottom of the first page of the table. If the use of the item is described in any part of this manual, reference to the applicable paragraph will suffice. If a separate manual exists covering the use of the item, a reference to the manual shall be included. Use foot notes, or asterisks to identify manual references. If most items require a manual reference an additional column "REFERENCE" may be added or manual reference may be included under part number.

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Consumable Materials Required

Item	Part Number	Specification	Nomenclature	Use and Application				
				A	B	C	D	E
				X			X	
					X			
						X		

3.2.1.3 Removal and Installation Procedures. In manuals requiring removal and installation procedures the removal procedures should be listed first. The standard maintenance procedures that apply to component removal and installation shall be summarized in the paragraph immediately preceding the first installation procedure. (See figure 5.) This summary shall include, but is not limited to, such items as applicable safety procedures, installation of protective covers, standard torque requirements, safety wire, determination of serviceability of replacement parts, and maintenance documentation requirements. The procedure shall be in sufficient detail, supported by illustrations, to enable removal and installation of each line replaceable unit in the system being covered. Illustrations will identify items to be connected or disconnected and attaching hardware to be installed or removed. Index numbers shall be used in the procedural text to reference the illustrated items for ease of identification. Each installation procedure shall include or reference the specific test or checkout procedure required to verify proper system operation after installation is completed. A reference to applicable intermediate level maintenance manual shall be included if such a manual exists.

3.2.1.4 Maintenance Procedures. In manuals requiring maintenance instructions the primary paragraph, MAINTENANCE, and subordinate paragraphs, INSPECTION, CLEANING, ADJUSTMENTS, LUBRICATION, OPERATIONAL CHECKOUT, and EXTREME CLIMATIC AND ENVIRONMENTAL INFORMATION, are mandatory. In cases where no requirements exist for one or more of the subordinate paragraphs the paragraph heading shall be included with a statement that no requirement exists.

3.2.1.4.1 Inspection. This paragraph shall include the procedure for accomplishing all inspection requirements specified in the applicable inspection manual (-6 series) for which the procedure is not obvious. This shall include obtaining access, inspection parameters, action required on defective components, and button-up after inspection is completed. The frequency of the inspection shall not be included as this is established by the inspection manual.

3.2.1.4.2 Cleaning. This paragraph shall include cleaning procedures for the equipment or system being covered. If standard cleaning procedures are adequate a reference to the applicable standard manual is acceptable. A reference to the tables containing the listing of materials and equipment required for cleaning is required when a specific cleaning procedure is provided.

3.2.1.4.3 Adjustments. This paragraph shall include all systems level adjustments required for equipments and systems installed in the RPV. Adjustment procedures shall include access provisions, adjustment methods and parameters, tests required to verify proper adjustment and equipment/system button-up. Adjustments required incidental to other maintenance/installation actions that are included as part of the other procedures shall be referenced if they could be required as a separate function.

3.2.1.4.4 Lubrication. This paragraph will include only those lubrication requirements incidental to maintenance actions. The normal RPV periodic lubrication is covered in the ground handling, servicing and airframe maintenance manual. In cases where no lubrication requirements exist a statement to that affect shall be included. Reference to the ground handling, servicing, and

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air frame maintenance manual or other manual containing lubrication instructions is acceptable.

3.2.1.4.5 Operational Checkout. Operational checkout/tests required to verify proper system/equipment operation following maintenance shall be identified. The checkout/test procedure shall be provided in this paragraph. If procedures are contained in the operational systems test manual or other maintenance manual the applicable manual shall be referenced in lieu of duplication of test procedures. In all cases the specific test requirements shall be identified.

3.2.1.4.6 Extreme Climatic and Environmental Information. This paragraph is mandatory for all manuals requiring this information. This paragraph shall include special maintenance, servicing, inspection, and lubrication requirements for maintenance of the system/equipment in extreme cold, extreme heat, rain and dust storms, high humidity, etc. If no added requirements exist a statement to that affect shall be included.

3.2.1.5 Servicing. This paragraph is mandatory in all manuals requiring maintenance and servicing actions. Since normal servicing is covered in the Ground Handling, Servicing, and Airframe Maintenance manual, only servicing incidental to maintenance actions shall be included in this paragraph. If no servicing requirement exists a statement to that affect shall be included. Reference to the Ground Handling, Servicing, and Airframe Maintenance manual is acceptable.

3.2.1.6 Troubleshooting. This section shall include detailed inspection and troubleshooting information to aid in identification, location, and correction of system/equipment malfunction. The method used for identification of system/equipment test points shall be explained. Reference to manual/section containing wiring, schematic, and cordage diagrams is acceptable. The primary paragraphs, INSPECTION and SYSTEM ANALYSIS, are mandatory for manuals requiring a troubleshooting section. The requirements and methods of determining defects through visual inspection shall be explained. Systems analysis (troubleshooting tables) shall be arranged to make maximum use of the purpose/ diagnosis column of the systems tests tables contained in the Operational Systems Test manuals. Troubleshooting tables shall be in the following format:

Step No.	Symptom	Probable Cause	Corrective Action

3.3 Detailed Outlines and Individual Manual Content Requirements. The following paragraphs define the content of the individual organizational (flight line) maintenance manuals specified herein. Equipment listings are for general guidance only and similar equipment should be added or deleted to individual manuals as required. Detailed outlines for individual manuals shall be prepared when directed by the procuring activity.

3.3.1 RPV General Manual.

3.3.1.1 General Content. This manual shall consist of a general but factual overall description of the RPV to provide a general knowledge to maintenance personnel. Normally no specific maintenance instructions shall be included in this manual except for classified information as defined in paragraph 3.3.1.2. Physical and functional descriptions shall be briefly written, appropriate manuals referenced, and supported as required by illustrations.

3.3.1.2 Classified Information. At the direction of the procuring activity, classified information related to maintenance instructions contained in other organizational (flight line) maintenance manuals may be included as added sections

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in the RPV General manual to permit remaining manuals to be published unclassified. Small amounts of classified information should be published in accordance with paragraph 3.1.9.

3.3.1.3 Arrangement. The RPV General manual shall be arranged in the following manner. Section titles and primary paragraph headings are mandatory. Subordinate paragraph headings are representative and shall be arranged to depict the equipment and systems comprising the specific RPV covered by the manual. Common nomenclatures shall be used throughout.

3.3.1.3.1 Section I, General Information. This section shall consist of two primary sideheads; introduction and description.

3.3.1.3.1.1 Introduction. The introduction shall identify the RPV by type, model, and series. It shall include a brief statement of the purpose and mission of the RPV.

3.3.1.3.1.2 Description. The description shall consist of a brief overview of the RPV. It shall identify the major subsystems (i.e., airframe, power plant, guidance, and payload) which comprise the RPV.

3.3.1.3.2 Section II, General Arrangement. This section shall consist of three primary sideheads; arrangement, dimensions and access and inspection features.

3.3.1.3.2.1 Arrangement. The arrangement of the RPV shall identify all major components of the airframe and RPV systems. Supporting illustrations shall be indexed to reflect each major component. Nomenclatures used in identification of components shall be consistent with nomenclatures reflected on applicable engineering drawings.

3.3.1.3.2.2 Dimensions. The overall dimensions of the RPV (i.e., length, wing span, height, sweepback, etc.) shall be provided. Principle dimensions shall be reflected on a line drawing of the overall RPV. The capacities of RPV systems (i.e., fuel, oil, coolant, etc.) shall be provided in tabular format arranged to identify the system, type of fluid, and capacity. The overall weight shall be provided for each authorized flight configuration. Empty and full weights for each configuration shall be provided.

3.3.1.3.2.3 Access and Inspection Features. Access and inspection features and openings shall be identified and illustrated. Components accessible through each opening shall be listed.

3.3.1.3.3 Section III, Operational Systems. This section shall identify and provide a brief description of each operational system comprising the RPV. The section shall consist of two primary sideheads; physical description and functional description.

3.3.1.3.3.1 Physical Description. Each operational system shall be identified and a brief physical description shall be provided. The description shall be written to the familiarization level to enable personnel to identify the overall physical design of the RPV.

3.3.1.3.3.2 Functional Description. A brief functional description of each system shall be provided. The description shall be written to the familiarization level to enable personnel to understand system level functions and the interrelationship between operational systems and total RPV operational parameters.

3.3.1.4 Manual Outline. The following is a sample outline of the RPV General manual. Primary sidehead are mandatory for this manual. Subordinate sideheads shall be dependent upon systems/equipment included in the RPV being covered in the manual.

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RPV GENERAL MANUAL

NOTE

Front matter shall be arranged as required by paragraph 3.1.8 of this specification.

SECTION I GENERAL INFORMATION

GENERAL

INTRODUCTION

PURPOSE

MISSION CAPABILITIES

DESCRIPTION

AIRFRAME

POWER PLANT

GUIDANCE

PAYLOAD

SECTION II GENERAL ARRANGEMENT

GENERAL

ARRANGEMENT

AIRFRAME

SYSTEMS

DIMENSIONS

OVERALL RPV

WINGS

STABILIZER

FUSELAGE

ACCESS AND INSPECTION FEATURES

ACCESS PANELS AND DOORS

EQUIPMENT ACCESS

EXTERNAL PODS AND EQUIPMENT

SECTION III OPERATIONAL SYSTEMS

GENERAL

PHYSICAL DESCRIPTION

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RPV GENERAL MANUAL (Continued)

SECTION III OPERATIONAL SYSTEMS

AIRFRAME

POWER PLANT

FLIGHT CONTROL

ENVIRONMENTAL, FUEL, AND OIL

ELECTRICAL

ELECTRONIC AND GUIDANCE

ARMAMENT

ELECTRONIC COUNTERMEASURES

RECONNAISSANCE

MISCELLANEOUS UTILITIES

RECOVERY

FUNCTIONAL DESCRIPTION

AIRFRAME

POWER PLANT

FLIGHT CONTROL

ENVIRONMENTAL, FUEL, AND OIL

ELECTRICAL

ELECTRONIC AND GUIDANCE

ARMAMENT

ELECTRONIC COUNTERMEASURES

RECONNAISSANCE

MISCELLANEOUS UTILITIES

RECOVERY

3.3.2 Ground Handling, Servicing, and Airframe Maintenance Manual.

3.3.2.1 General Content. This manual shall contain all essential organizational (flight line) maintenance instructions for ground handling and servicing the RPV, and for maintenance of such airframe components and systems or equipments as do not properly fall under the categories of structural repair.

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3.3.2.2 Arrangement. The Ground Handling, Servicing, and Airframe Maintenance manual shall be arranged in the following manner. Section titles and primary paragraph headings are mandatory. Subordinate paragraph headings are representative and shall be arranged to depict the equipment and systems comprising the specific RPV covered by the manual.

3.3.2.2.1 Section I, Ground Handling. This section shall provide all required instructions for safe ground handling of the RPV by organizational maintenance personnel. All required equipment shall be identified and use explained. Procedures shall include but are not limited to roll transfer, towing, hoisting, packing, storage, and general handling during maintenance. The primary paragraph headings shall consist of equipment required, roll transfer, towing, packing, storage, and handling during maintenance. Additional primary headings may be added to cover any ground handling functions peculiar to the RPV being covered.

3.3.2.2.1.1 Equipment Required. All equipment and materials required for ground handling shall be included as required by paragraph 3.2.1.2 of this specification.

3.3.2.2.1.2 Roll Transfer. Procedures for performing roll transfer from trailer to trailer and trailer to workstand shall be provided.

3.3.2.2.1.3 Towing. Towing procedures shall include preparation for towing, towing vehicle hookup, towing operation, towing speeds, safety precautions and personnel requirements. Special requirements for towing at night and during inclement weather (high winds and snow) shall be included.

3.3.2.2.1.4 Parking. Procedures shall be provided for parking the RPV inside an enclosure and outside an enclosure. Mooring procedures shall be included. Special requirements for parking and mooring during inclement weather (high winds and snow) shall be provided. Include requirements for covers, locks, etc. Personnel and equipment safety shall be emphasized.

3.3.2.2.1.5 Storage. Procedures for limited storage considered to be outside the parameters of the RPV storage manual shall be provided. In cases where all requirements are contained in the storage manual then a reference to that manual is acceptable.

3.3.2.2.1.6 Handling During Maintenance. Handling procedures related to the RPV during maintenance and checkout shall be provided. Grounding, protective coverings, steps, pushpoints, and grips shall be included. Include special handling procedures and precautions related to maintenance of a partially assembled RPV. Type and quantity of fire protection equipment required and equipment positioning shall be included.

3.3.2.2.2 Section II, Servicing. This section shall include all required instructions for safe servicing of the RPV systems by organizational maintenance personnel. All required equipment shall be identified and use explained. Procedures shall include but are not limited to servicing of RPV fuel, oil, and cooling systems, battery and sump (vent) jar servicing, and service requirements for RPV systems/components (i.e., pitot-static system, engine, recovery system, etc.). Additional primary headings may be added to this section to cover servicing of systems/components peculiar to the RPV being covered by this manual.

3.3.2.2.2.1 Equipment Required. All equipment and materials required for servicing of RPV systems shall be included as required by paragraph 3.2.1.2 of this specification.

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3.3.2.2.2.2. Fuel. Procedures for filling, draining, and purging the RPV fuel system shall be provided. Fuel filter servicing shall be included. Special emphasis shall be placed on safety precautions, fire protection, and hazard climination during servicing of the RPV fuel system and components.

3.3.2.2.2.3 Lubricating Oil. Procedures for filling, draining, flushing, and priming the lubricating oil system shall be provided. Special emphasis shall be placed on safety precautions, fire protection, and hazard elimination during servicing of the RPV lubricating oil system.

3.3.2.2.2.4 Engine. Procedures for servicing engine systems and components shall be provided. Periodic servicing of engine filters and components shall be included. Servicing of engine components incident to maintenance shall not be included, but shall be included in the applicable maintenance procedure in the Power Plant manual.

3.3.2.2.2.5 Batteries. Servicing procedures for all RPV batteries (i.e., main battery, recovery batteries, beacon batteries, programmer batteries, etc.) shall be provided. Servicing of batter vent (sump) jars and vent systems shall be included.

3.3.2.2.2.6 Cooling Systems. Procedures for filling, draining, and purging of RPV coolant systems (i.e., ammonia, Freon, etc.) shall be provided. Cleaning and servicing of cooling system filters and vents shall be included. Safety procedures shall be emphasized when procedure involves servicing of hazardous materials/liquids (i.e., ammonia).

3.3.2.2.2.7 Pitot-Static System. Procedures for locating pitot-static drains, opening drain ports, draining condensation, securing drain ports, and leak testing the pitot-static system shall be provided.

3.3.2.2.2.8 Recovery System. Procedures for servicing recovery system components (i.e., release chambers, swivel release, etc.) shall be provided.

3.3.2.2.3 Section III, Lubrication. This section shall identify all lubrication requirements for RPV systems and components. Equipment required for lubrication, lubricants required, and lubricant application methods shall be provided. This section shall consist of three primary paragraph headings; equipment required, lubricant list, and lubricant application.

3.3.2.2.3.1 Equipment Required. All equipment and materials required to accomplish lubrication of RPV systems and components shall be identified as required by paragraph 3.2.1.2 of this specification.

3.3.2.2.3.2 Lubricant List. All lubricants required for RPV lubrication shall be identified in tabular format as follows:

Item	Nomenclature	Identification Number or specification	Alternate	Use or Purpose

3.3.2.2.3.3 Lubricant Application. All RPV lubrication requirements, frequency, and method of application shall be provided in chart form or by using supporting illustration. Lubrication points shall be illustrated. Lubrication required incident to maintenance tasks shall not be included but shall be included in the maintenance procedure in the applicable system maintenance manual.

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3.3.2.2.4 Section IV, Airframe Maintenance. This section shall identify all equipment and materials required to accomplish airframe maintenance procedures and complete testing after maintenance. Leading particulars shall be tabulated and separate physical and functional descriptions of airframe components provided. Procedures for removal and installation of airframe components shall be included. Procedures for inspection, cleaning, rigging, adjustment, and lubrication are required. Maintenance of fairings, doors, panels, and structures to include replacement of seals, fasteners, and connectors and surface treatment shall be included. Primary paragraph headings listed in the manual outline are mandatory. Additional headings may be added to cover maintenance of airframe components peculiar to the RPV being covered by the manual. Procedures for RPV leveling, alignment, weight and balance and troubleshooting airframe components are required.

3.3.2.2.4.1 Equipment Required. All equipment and materials required to accomplish airframe maintenance procedures shall be identified as required by paragraph 3.2.1.2 of this specification.

3.3.2.2.4.2 Description and Leading Particulars. A separate physical and functional description of the airframe and major components shall be provided. Airframe leading particulars shall be tabulated.

3.3.2.2.4.3 Removal and Installation. Removal and installation procedures shall be prepared as outlined in paragraph 3.2.1.3 of this specification. Subordinate paragraph headings contained in the manual outline are representative. Actual paragraph headings shall be dependent upon components comprising the RPV being covered by the manual.

3.3.2.2.4.4 Maintenance. Maintenance procedures shall be prepared as required by paragraph 3.2.1.4 of this specification. Additional subordinate paragraphs shall be added for airframe components peculiar to the RPV being covered by the manual.

3.3.2.2.4.5 Leveling and Alignment. Procedures for leveling the RPV, determining wing incidence and twist, and leveling and checking alignment of the vertical stabilizer shall be provided. All required limitations and tolerances shall be included.

3.3.2.2.4.6 Weight and Balance. Procedures for determining actual weight and center of gravity of the RPV shall be provided. Each approved method (i.e., graphic and analytical) shall be included. Procedures for removal and installation of ballast plates shall be provided. Ballast locations and limitations shall be included. Applicable ballast plates shall be identified and weights for each shall be specified.

3.3.2.2.4.7 Troubleshooting. Troubleshooting procedures shall be provided as required by paragraph 3.2.1.7 of this specification.

3.3.2.3. Manual Outline. The following is a sample outline of the Ground Handling, Servicing, and Airframe Maintenance manual. Primary sideheads are mandatory. Those subordinate sideheads required by preceding paragraphs are mandatory. Additional primary and subordinate paragraph headings may be added for equipment peculiar to the RPV being covered by the manual (i.e., landing gear, pneudraulics, etc.).

GROUND HANDLING, SERVICING, AND AIRFRAME MAINTENANCE MANUAL

NOTE

Front matter shall be arranged as required by paragraph 3.1.8 of this specification.

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SECTION I GROUND HANDLING

GENERAL

EQUIPMENT REQUIRED

TRAILERS

Transportation

Positioning

WORKSTANDS

ADAPTERS

Fuselage

Nacelle

Empennage

Rotating

Pod

Extension

Wing

HOISTING EQUIPMENT

Gantry Hoist

Lifting Beams

TOWING EQUIPMENT

Towing Vehicle

Tow Bar and Extensions

TRANSFER

PREPARATION

TRANSFER

ROLL

HOISTING

TOWING

PREPARATION

TOW PROCEDURE

PARKING

INSIDE ENCLOSURE

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GROUND HANDLING, SERVICING, AND AIRFRAME MAINTENANCE MANUAL (Continued)

SECTION I (Continued)

OUTSIDE

MOORING

STORAGE

TEMPORARY

EXTENDED

HANDLING DURING MAINTENANCE

GROUNDING REQUIREMENTS

PROTECTIVE COVERINGS

STEPS, PUSHPOINTS AND GRIPS

FIRE PROTECTION

SECTION II SERVICING

GENERAL

EQUIPMENT REQUIRED

SAFETY EQUIPMENT

Fire Extinguishers

Squib Shorting Kit

Safety Pins

Grounding Cables

Safety Containers

Safety Placards

PROTECTIVE COVERS

SERVICING EQUIPMENT

Fuel Truck/Container

Oil Container

Funnels

Latch Pressure Gage

Purge and Pressurization Cart

FUEL

PREPARATION

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GROUND HANDLING, SERVICING, AND AIRFRAME MAINTENANCE MANUAL (Continued)

SECTION II (Continued)

FUELING

DEFUELING

PURGING

FILTERS

LUBRICATING OIL

FILLING

DRAINING

FLUSHING TANK

PRIMING

ENGINE

COMPRESSOR DISCHARGE FILTER

MAGNETIC PLUG

BATTERIES

MAIN BATTERY

RECOVERY BATTERY

LOCATOR BEACON BATTERIES

PROGRAMMER BATTERY PACK

BATTERY VENT (SUMP) JAR

COOLING SYSTEMS

AMMONIA

FREON

TWT COLLANT

PITOT-STATIC SYSTEM

DRAINING

LEAK TEST

RECOVERY SYSTEM

CHAMBERS

GROUND RELEASE SWIVEL

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GROUND HANDLING, SERVICING, AND AIRFRAME MAINTENANCE MANUAL (Continued)

SECTION III LUBRICATION

GENERAL

EQUIPMENT REQUIRED

LUBRICANT LIST

LUBRICANT APPLICATION

SECTION IV AIRFRAME MAINTENANCE

GENERAL

EQUIPMENT REQUIRED

TEST EQUIPMENT

TOOLS

CONSUMABLE MATERIALS

DESCRIPTION AND LEADING PARTICULARS

PHYSICAL DESCRIPTION

Fuselage

Wings

Empennage

FUNCTIONAL DESCRIPTION

Fuselage

Wings

Empennage

LEADING PARTICULARS

REMOVAL

WING

Wing Tips

EMPENNAGE

CONTROL SURFACES

Ailerons

Elevators

Rudder

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GROUND HANDLING, SERVICING, AND AIRFRAME MAINTENANCE MANUAL (Continued)

SECTION IV (Continued)

NOSE CONE

NOSE MODULE

TAIL CONE

NACELLE

Nacelle Skid

NOSE COWL

EQUIPMENT COMPARTMENT DOORS

Forward

Aft (Main)

FAIRINGS AND ACCESS PANELS

TUBING AND LINES

PRESSURE LATCHES

BULKHEAD FEED-THRU CONNECTORS

Pressurized

Unpressurized

EQUIPMENT TRAYS AND RACKS

PODS

PYLONS

INSTALLATION

WING

Wing Tips

EMPENNAGE

CONTROL SURFACES

Ailerons

Elevators

Rudders

NOSE CONE

NOSE MODULE

TAIL CONE

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GROUND HANDLING, SERVICING, AND AIRFRAME MAINTENANCE MANUAL (Continued)

SECTION IV (Continued)

NACELLE

Nacelle Skid

NOSE COWL

EQUIPMENT COMPARTMENT DOORS

Forward

Aft (Main)

FAIRINGS AND ACCESS PANELS

TUBING AND LINES

PRESSURE LATCHES

BULKHEAD FEED-THRU CONNECTORS

Pressurized

Unpressurized

EQUIPMENT TRAYS AND RACKS

RODS

3.3.3 Uncrating, Crating, Assembly, and Disassembly Manual.

3.3.3.1 General Content. This manual shall contain all instructions necessary for uncrating, crating, assembly, and disassembly of the RPV by organizational maintenance personnel.

3.3.3.2 Arrangement. The Uncrating, Crating, Assembly, and Disassembly manual shall be arranged in the following manner. Section titles and primary paragraph heading are mandatory. Subordinate paragraph headings are representative and shall be arranged to depict the requirements and procedures for the RPV being covered by the manual.

3.3.3.2.1 Section I, Introduction and General Information. This section shall identify the packing methods, shipping containers and container contents. Inspection and inventory requirements shall be provided. Special requirements and precautions related to maintenance under extreme climatic conditions. Primary paragraph headings shall consist of introduction, container identification, inspection, inventory, extreme climatic and environmental information and storage. Subordinate paragraphs may vary depending on equipment arrangement of RPV being covered by the manual.

3.3.3.2.1.1 Introduction. The introduction shall identify the packing methods for the RPV. A brief overview of situations requiring assembly, partial disassembly, and complete disassembly shall be included.

3.3.3.2.1.2 Container Identification. All shipping containers (i.e., skids, racks, cartons, etc.) shall be identified and illustrated. The application of each shall be explained.

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3.3.3.2.1.3 Inspection. The requirements for accomplishing inspection of newly received RPV and those being prepared for shipment shall be provided. The inspections in this section are limited to inspection of completely crated RPV. The inspections performed as part of uncrating and crating shall be covered in section II.

3.3.3.2.1.4 Inventory. Inventory of crated RPV as they are received or shipped shall be provided. Inventory of container contents shall be covered in section II as part of uncrating and crating.

3.3.3.2.1.5 Extreme Climatic and Environmental Information. Extreme climatic and environmental information shall be provided as required in paragraph 3.2.1.4.6 of this specification.

3.3.3.2.1.6 Storage. Identify and peculiar requirements necessary when crating an RPV for short or long term storage and when uncrating an RPV that was removed from storage.

3.3.3.2.2 Section II, Uncrating and Crating. This section shall consist of four primary paragraph headings; equipment required, uncrating, crating, and forms and documentation. Procedures shall be included for accomplishing required inspections, inventory, preservation, packing, marking, and documentation. Subordinate paragraphs shall be dependent upon equipment complexity but in all cases shall include information required by this paragraph.

3.3.3.2.2.1 Equipment Required. All equipment and materials required to accomplish uncrating and crating procedures shall be identified as required by paragraph 3.2.1.2 of this specification. Procedures for construction of shipping skids/racks shall be provided when applicable. A table of packing containers and materials shall be provided.

3.3.3.2.2.2 Uncrating. Uncrating procedures shall include preparation, inspections required before and during uncrating, inventory before and during uncrating, and disposition of packing materials and containers.

3.3.3.2.2.3 Crating. Crating procedures shall include preparation, inspections required before, during and after crating, inventory, preservation, cleaning of components, marking and sealing of containers.

3.3.3.2.2.4 Forms and Documentation. Forms required for inspection, inventory, and preservation shall be identified and documentation procedures provided. All phases of uncrating and crating shall be included in documentation procedures (i.e., receiving, shipping, storage, etc.).

3.3.3.2.3 Section III, Assembly and Disassembly. This section shall consist of at least three primary paragraph headings; equipment required, assembly, and disassembly. Procedures shall be included for assembly and disassembly of the complete RPV. Subordinate paragraphs shall be dependent upon equipment/systems comprising the specific RPV being covered by the manual.

3.3.3.2.3.1 Equipment Required. All equipment and materials required to accomplish complete assembly and disassembly of the RPV shall be identified as required by paragraph 3.2.1.2 of this specification.

3.3.3.2.3.2 Assembly. Procedures for RPV assembly shall include installation of all components necessary to fully assemble a crated RPV. When installation procedures for a component are included in another manual (i.e., recovery system, power plant, etc.) a reference to the applicable manual will suffice. Assembly procedures shall include applicable inspections, rigging, adjustments, lubrication, servicing, and documentation requirements.

3.3.3.2.3.3 Disassembly. Procedures for RPV disassembly shall include removal

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of all components necessary to disassemble a RPV for crating and packing. When removal procedures for a component are included in another manual (i.e., recovery system, power plant, etc.) a reference to the applicable manual will suffice. Disassembly procedures shall include applicable inspections, adjustments, lubrication, servicing, cleaning, and documentation.

3.3.3.2.4 Section IV, Loose Parts Inventory. This section shall consist of at least three primary paragraph headings; inventory listing, reporting errors, replacement items.

3.3.3.2.4.1 Inventory Listing. A listing of loose parts (i.e., attaching hardware, rod ends, etc) required to complete assembly of the RPV shall be provided in tabular format. Items in each container will be listed in numerical sequence by part number. Alternate items shall be listed when applicable. Information contained in the RPV Inventory manual shall be referenced and not duplicated.

3.3.3.2.4.2 Replacement Items. Procedures to enable maintenance personnel to obtain replacement components for items found missing or defective shall be included. This shall include items under contractors warranty and items to be obtained through normal supply channels.

3.3.3.3 Manual Outline. The following is a sample outline of the Uncrating, Crating, Assembly, and Disassembly manual. Primary sideheads listed are mandatory. Additional primary paragraph headings may be added to cover peculiar equipment or requirements for the RPV covered by the manual.

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UNCRATING, CRATING, ASSEMBLY, AND DISASSEMBLY MANUAL

NOTE

Front matter shall be arranged as required by paragraph 3.1.8 of this specification.

SECTION I INTRODUCTION AND GENERAL INFORMATION

GENERAL

INTRODUCTION

PACKING METHODS

Factory Packing

Field Packing

ASSEMBLY

DISASSEMBLY

CONTAINER IDENTIFICATION

FUSELAGE SKID

WING RACK

STABILIZER RACKS

Vertical

Horizontal

SHIPPING CARTONS

INSPECTION

RECEIVING

SHIPPING

INVENTORY

RECEIVING

SHIPPING

Air Shipment

Surface Shipment

EXTREME CLIMATIC AND ENVIRONMENTAL INFORMATION

STORAGE REQUIREMENTS

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UNCRATING, CRATING, ASSEMBLY, AND DISASSEMBLY MANUAL (Continued)

SECTION II UNCRATING AND CRATING

GENERAL

EQUIPMENT REQUIRED

HANDLING EQUIPMENT

TOOLS

CONSUMABLE MATERIALS

PACKING CONTAINERS AND MATERIALS

UNCRATING

PREPARATION

INVENTORY OF CONTAINER CONTENTS

INSPECTION

UNCRATING PROCEDURE

DISPOSITION OF PACKING MATERIALS

CRATING

PREPARATION

INSPECTION

PRESERVATION

INVENTORY

CRATING PROCEDURE

MARKING REQUIREMENTS

FORMS AND DOCUMENTATION

RECEIVING

SHIPPING

STORAGE

SECTION III ASSEMBLY AND DISASSEMBLY

GENERAL

EQUIPMENT REQUIRED

HANDLING EQUIPMENT

TEST EQUIPMENT

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UNCRATING, CRATING, ASSEMBLY, AND DISASSEMBLY MANUAL (Continued)

SECTION III (Continued)

TOOLS

CONSUMABLE MATERIALS

ASSEMBLY

PREPARATION

FUSELAGE ASSEMBLY

WING INSTALLATION

- Wing Tips
- Wing Ballast/Flutter Booms
- Fairings

EMPENNAGE INSTALLATION

- Parachute Container
- Fairings
- Shear Plates

NOSE CONE INSTALLATION

PYLON INSTALLATION

- Fairings

EXTERNAL PODS

OPERATIONAL CHECKOUT

INSPECTION

ADJUSTMENTS

LUBRICATION

SERVICING

FORMS AND DOCUMENTATION

DISASSEMBLY

PREPARATION

- Defueling
- Purge
- Oil and Coolant
- Batteries
- Explosives

PYLON REMOVAL

EXTERNAL PODS

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UNCRATING CRATING, ASSEMBLY, AND DISASSEMBLY MANUAL (Continued)

SECTION III (Continued)

NOSE CONE REMOVAL

EMPENNAGE REMOVAL

Parachute Containers

WING REMOVAL

Fairings

Wing Ballast/Flutter Booms

Wing Tips

FUSELAGE DISASSEMBLY

INSPECTION

FORMS AND DOCUMENTATION

SECTION IV LOOSE PARTS INVENTORY

GENERAL

INVENTORY LISTING

REPORTING ERRORS

REPLACEMENT ITEMS

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3.3.4 Power Plant Manual.

3.3.4.1 General Content. This manual shall describe the RPV power plant and provide engine buildup/tear down instructions, procedures for removal and installation of the power plant and components, and maintenance, servicing, and troubleshooting instructions.

3.3.4.2 Section I, Introduction and Description. This section shall consist of at least four primary paragraph headings; introduction, physical description, functional description, and engine limitations. An introduction to the power plant and systems, a complete physical and functional description, and engine operating limits (i.e., egt, rpm, throttle settings, etc.) shall be provided.

3.3.4.2.1 Introduction. The introduction shall identify the specific type/model engine used in the RPV and list the major components comprising the power plant. Power plant leading particulars shall be tabulated.

3.3.4.2.2 Physical Description. A complete physical description of the engine, accessories, engine mounts, engine fuel, oil, pneumatic, electrical, instrument, ignition and exhaust systems shall be provided. Descriptions shall include physical characteristics, dimensions, weights, locations, design features, and mounting provisions.

3.3.4.2.3 Functional Description. A complete functional description of the power plant and power plant systems shall be provided. Functional interface with RPV systems/equipment shall be included.

3.3.4.2.4 Engine Limitations. Engine operating limits (i.e., fuel specifications, throttle settings, rpm limits, egt limits, etc.) shall be tabulated. Maintenance actions required when limits are exceeded shall be identified.

3.3.4.3 Section II, Removal and Installation. This section shall consist of at least six primary paragraph headings; equipment required, engine buildup/teardown, engine removal, engine installation, component removal and component installation. Procedures shall include preparation requirements as well as step-by-step procedures for all removal and installation tasks.

3.3.4.3.1 Equipment Required. All equipment and materials required to accomplish removal/installation tasks shall be identified as required by paragraph 3.2.1.2 of this specification.

3.3.4.3.2 Engine Buildup/Tear Down. Procedures for installation of accessories, mounts, and other components required to prepare a bare engine for installation in the RPV shall be provided. Procedures for removal of all components shall be provided. Items to be retained for reuse shall be identified.

3.3.4.3.3 Engine Removal. Procedures for preparation and removal of the power plant from the RPV shall be provided.

3.3.4.3.4 Engine Installation. Procedures for preparation and installation of the power plant in the RPV shall be provided.

3.3.4.3.5 Component Removal. Procedures for removal of components from the power plant while installed in the RPV shall be provided. When procedures are identical to those provided in engine buildup/tear down instructions (paragraph 3.3.4.3.2 of this specification) a reference to the procedures will suffice.

3.3.4.3.6 Component Installation. Procedures for installation of components of the power plant while installed in the RPV shall be provided. When procedures are identical to those provided in engine buildup/tear down instructions (paragraph 3.3.4.3.2 of this specification) a reference to the procedures will suffice. Procedures for removal and installation of lines, fittings, and wiring harnesses may be included as a combined task immediately following the last installation procedure.

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3.3.4.4 Section III, Maintenance and Servicing. This section shall consist of at least three primary paragraph headings; equipment required, maintenance, and servicing.

3.3.4.4.1 Equipment Required. All equipment and materials required to accomplish power plant maintenance and servicing tasks shall be identified as required by paragraph 3.2.1.2 of this specification.

3.3.4.4.2 Maintenance. Procedures for accomplishing required maintenance tasks for the power plant and power plant systems shall be provided. Procedures shall include the following subordinate paragraph headings; inspection, cleaning, adjustment, lubrication, operational checkout, and extreme climatic and environmental information. Additional subordinate paragraphs may be added to provide maintenance procedures for equipment peculiar to the power plant and RPV being covered by the manual. A reference to preservation/storage procedures shall be included.

3.3.4.4.3 Servicing. Servicing required incident to maintenance tasks shall be included. A reference to the normal service procedures in the ground handling, servicing, and airframe maintenance manual shall be included as applicable.

3.3.4.5 Section IV, Troubleshooting. This section shall consist of at least two primary paragraph headings; inspection and system analysis.

3.3.4.5.1 Inspection. Procedures for visual inspection of components to be accomplished as part of troubleshooting shall be provided as required by paragraph 3.2.1.6 of this specification.

3.3.4.5.2 System Analysis. Trouble analysis as required by paragraph 3.2.1.6 of this specification shall be provided.

3.3.4.6 Manual Outline. The following is a sample outline of the Power Plant manual. Primary sideheads are mandatory. Subordinate sideheads may require rearrangement or additional paragraphs may be required to include specific equipment comprising the power plant of the RPV being covered by the manual.

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POWER PLANT MANUAL

NOTE

Front matter shall be arranged as required by paragraph 3.1.8 of this specification.

SECTION I INTRODUCTION AND DESCRIPTION

GENERAL

INTRODUCTION

POWER PLANT

LEADING PARTICULARS

PHYSICAL DESCRIPTION

ENGINE

- Compressor Section
- Accessory Drive
- Combustor Section
- Turbine Section
- Exhaust Duct

ENGINE MOUNTS

- Forward Mounts
- Aft Mounts

ENGINE FUEL SYSTEM

- Fuel Control
- Starting Fuel Valve
- Fuel Nozzles
- Fuel Filters
- Fuel Lines and Fittings

ENGINE OIL SYSTEM

- Oil Pump
- Oil Filter
- Oil Cooler
- Magnetic Plug

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POWER PLANT MANUAL (Continued)

SECTION I (Continued)

ENGINE OIL SYSTEM

Oil Pump

Oil Filter

Oil Cooler

Magnetic Plug

Oil Lines and Fittings

ENGINE PNEUMATIC SYSTEM

Compressor Discharge Lines and Fittings

Compressor Discharge Filter

ENGINE ELECTRICAL SYSTEM

Starter-Generator

External Power Receptacle

Wiring Harnesses

ENGINE INSTRUMENT SYSTEM

Tachometer-Generator

Thermocouples and Leads

Oil Pressure Transducer

Fuel Flow Transducer

ENGINE IGNITION SYSTEM

Ignitors

Ignitor Leads

ENGINE EXHAUST SYSTEM

Tailpipe

Tailpipe Restrictors

Exhaust Shrouds

FUNCTIONAL DESCRIPTION

ENGINE

Air/Gas Flow

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POWER PLANT MANUAL (Continued)

SECTION I (Continued)

ENGINE FUEL SYSTEM

ENGINE OIL SYSTEM

ENGINE PNEUMATIC SYSTEM

ENGINE ELECTRICAL SYSTEM

ENGINE INSTRUMENT SYSTEM

ENGINE IGNITION SYSTEM

ENGINE EXHAUST SYSTEM

ENGINE LIMITATIONS

FUEL CONTROL

SPEED LIMITATIONS

EGT LIMITATIONS

SECTION II REMOVAL AND INSTALLATION

GENERAL

EQUIPMENT REQUIRED

TEST EQUIPMENT

TOOLS

HANDLING EQUIPMENT

CONSUMABLE MATERIALS

ENGINE BUILDUP/TEAR DOWN

BARE ENGINE TESTS

QUICK ENGINE CHANGE ITEMS

QUICK ENGINE CHANGE KIT INSTALLATION

QUICK ENGINE CHANGE KIT REMOVAL

ENGINE REMOVAL

PREPARATION FOR ENGINE REMOVAL

NACELLE REMOVAL AND/OR FUSELAGE DISCONNECT

ENGINE REMOVAL FROM NACELLE/AIRFRAME

ENGINE INSTALLATION

PREPARATION FOR ENGINE INSTALLATION

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POWER PLANT MANUAL (Continued)

SECTION II (Continued)

ENGINE INSTALLATION IN NACELLE

NACELLE INSTALLATION AND/OR FUSELAGE CONNECTIONS

COMPONENT REMOVAL

PREPARATION

ENGINE MOUNTS

Forward Mounts

Aft Mounts

FUEL CONTROL AND THROTTLE ACTUATOR

STARTING FUEL VALVE

FUEL NOZZLES

FUEL FILTER

FUEL LINES AND FITTINGS

OIL FILTER

OIL COOLER

STARTING FUEL BOOST PUMP

INTERSTAGE BLEED VALVE

MAGNETIC PLUG

OIL LINES AND FITTINGS

COMPRESSOR DISCHARGE LINES AND FITTINGS

COMPRESSOR DISCHARGE FILTER

STARTER-GENERATOR

EXTERNAL POWER RECEPTACLE

WIRING HARNESSES

TACHOMETER-GENERATOR

THERMOCOUPLES AND LEADS

OIL PRESSURE TRANSDUCER

FUEL FLOW TRANSDUCER

IGNITERS

IGNITER LEADS

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POWER PLANT MANUAL (Continued)

SECTION II (Continued)

EXHAUST SHROUDS

TAILPIPE

COMPONENT INSTALLATION

PREPARATION

ENGINE MOUNTS

Forward Mounts

Aft Mounts

FUEL CONTROL AND THROTTLE ACTUATOR

STARTING FUEL VALVE

FUEL NOZZLES

FUEL FILTER

FUEL LINES AND FITTINGS

OIL FILTER

OIL COOLER

STARTING FUEL BOOST PUMP

INTERSTAGE BLEED VALVE

MAGNETIC PLUG

OIL LINES AND FITTINGS

COMPRESSOR DISCHARGE LINES AND FITTINGS

COMPRESSOR DISCHARGE FILTER

STARTER-GENERATOR

EXTERNAL POWER RECEPTACLE

WIRING HARNESSES

TACHOMETER-GENERATOR

THERMOCOUPLES AND LEADS

OIL PRESSURE TRANSDUCER

FUEL FLOW TRANSDUCER

IGNITERS

IGNITER LEADS

EXHAUST SHROUDS

TAILPIPE

SECTION III MAINTENANCE AND SERVICING

GENERAL

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POWER PLANT MANUAL (Continued)

SECTION III (Continued)

EQUIPMENT REQUIRED

MAINTENANCE

INSPECTION

CLEANING

ADJUSTMENT

LUBRICATION

Oil Pressure

Fuel Control

Throttle Actuator

Tailpipe Temperature

OPERATIONAL CHECKOUT

EXTREME CLIMATIC AND ENVIRONMENTAL INFORMATION

SERVICING

Fuel Filter

Oil Filter

CDP Filter

Magnetic Plug

Oil System Purge

Oil System Prime

SECTION IV TROUBLESHOOTING

GENERAL

INSPECTION

SYSTEM ANALYSIS

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3.3.5 Environmental, Fuel and Oil System Manual.

3.3.5.1 General Content. This manual shall contain all essential organizational (flight line) maintenance instructions for the RPV environmental, fuel and oil systems. Descriptions, component removal and installation, maintenance, servicing, and troubleshooting shall be included for each system.

3.3.5.2 Section I, Introduction and General Information. This section shall consist of at least two primary paragraph headings; introduction and leading particulars.

3.3.5.2.1 Introduction. The introduction shall identify the major components comprising the environmental, fuel and oil systems.

3.3.5.2.2 Leading Particulars. Leading particulars for each system shall be tabulated. Operating limitations, fluid specifications and system capacities shall be included.

3.3.5.3 Section II, Environmental Systems. This section shall consist of at least the following primary paragraph headings; physical description, functional description, equipment required, removal, installation, maintenance, servicing, and troubleshooting.

3.3.5.3.1 Physical Description. A complete physical description shall be provided for components comprising the environmental systems. RPV cooling systems, heating systems, and pressurization systems shall be described. The physical description shall include physical characteristics, dimensions, weights, locations, design features, and mounting methods.

3.3.5.3.2 Functional Description. A complete functional description, supported by functional block diagrams, shall be provided for each system. The functional description shall include system interface with the RPV or other RPV systems. Each cooling, heating, and pressurization system shall be described separately.

3.3.5.3.3 Equipment Required. All equipment and materials required to accomplish removal, installation, maintenance, servicing, and troubleshooting of the environmental systems shall be identified as required by paragraph 3.2.1.2 of this specification.

3.3.5.3.4 Removal. Procedures for removal of environmental system components from the RPV shall be provided as required by paragraph 3.2.1.3 of this specification.

3.3.5.3.5 Installation. Procedures for installation of environmental system components on the RPV shall be provided as required by paragraph 3.2.1.3 of this specification.

3.3.5.3.6 Maintenance. Maintenance procedures shall include inspection, cleaning, adjustments, lubrication, operational checkout, leak tests, and extreme climatic and environmental information per paragraph 3.2.1.4 of this specification.

3.3.5.3.7 Servicing. Servicing required incidental to maintenance shall be included per paragraph 3.2.1.5 of this specification.

3.3.5.3.8 Troubleshooting. Troubleshooting procedures shall be provided for the environmental systems as required by paragraph 3.2.1.6 of this specification.

3.3.5.4 Section III, Fuel System. This section shall consist of a least the following primary paragraph headings; physical description, functional description, equipment required, removal, installation, maintenance, servicing, and troubleshooting.

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3.3.5.4.1 Physical Description. A complete physical description shall be provided for components comprising the fuel system. The physical description shall include physical characteristics, dimensions, weights, locations, design features, and mounting methods.

3.3.5.4.2 Functional Description. A complete functional description, supported by functional block diagrams shall be provided for the fuel system. Fuel system interface with other RPV systems (i.e., power plant, electrical, etc.) shall be included.

3.3.5.4.3 Equipment Required. All equipment and materials required to accomplish removal, installation, maintenance, servicing, and troubleshooting of the RPV fuel system shall be identified as required by paragraph 3.2.1.2 of this specification.

3.3.5.4.4 Removal. Procedures for removal of fuel system components from the RPV shall be provided as required by paragraph 3.2.1.3 of this specification.

3.3.5.4.5 Installation. Procedures for installation of fuel system components on the RPV shall be provided as required by paragraph 3.2.1.3 of this specification.

3.3.5.4.6 Maintenance. Maintenance procedures shall include inspection, cleaning, adjustments, lubrication, operational checkout, leak tests, and extreme climatic and environmental information per paragraph 3.2.1.4 of this specification. Include SOAP sample procedures if required.

3.3.5.4.7 Servicing. Servicing required incidental to maintenance shall be included per paragraph 3.2.1.5 of this specification.

3.3.5.4.8 Troubleshooting. Troubleshooting procedures for the fuel system shall be provided as required by paragraph 3.2.1.6 of this specification.

3.3.5.5 Section IV, Oil System. This section shall consist of at least the following primary paragraph headings; physical description, functional description, equipment required, removal, installation, maintenance, servicing, and trouble-shooting.

3.3.5.5.1 Physical Description. A complete physical description shall be provided for components comprising the oil system. The physical description shall include physical characteristics, dimensions, weights, locations, design features, and mounting methods.

3.3.5.5.2 Functional Description. A complete functional description, supported by functional block diagrams, shall be provided for the oil system. Oil system interface with other RPV systems (i.e., power plant, airframe, etc.) shall be included.

3.3.5.5.3 Equipment Required. All equipment and materials required to accomplish removal, installation, maintenance, servicing, and troubleshooting of the RPV oil system shall be identified as required by paragraph 3.2.1.2 of this specification.

3.3.5.5.4 Removal. Procedures for removal of oil system components from the RPV shall be provided as required by paragraph 3.2.1.3 of this specification.

3.3.5.5.5 Installation. Procedures for installation of oil system components on the RPV shall be provided as required by paragraph 3.2.1.3 of this specification.

3.3.5.5.6 Maintenance. Maintenance procedures shall include inspection, cleaning, adjustments, lubrication, operational checkout, leak tests, and extreme climatic and environmental information as required by paragraph 3.2.1.4 of this specification.

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3.3.5.5.7 Servicing. Servicing required incidental to maintenance shall be included per paragraph 3.2.1.5 of this specification.

3.3.5.5.8 Troubleshooting. Troubleshooting procedures for the oil system shall be provided as required by paragraph 3.2.1.6 of this specification.

3.3.5.6 Manual Outline. The following is a sample outline of the Environmental, Fuel and Oil System manual. Primary sideheads are mandatory for this manual. Subordinate sideheads under maintenance and troubleshooting are mandatory. Remaining subordinate paragraphs shall be dependent upon equipment/systems comprising the specific RPV being covered by the manuals.

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ENVIRONMENTAL, FUEL AND OIL SYSTEM MANUAL

NOTE

Front matter shall be arranged as required by paragraph 3.1.8 of this specification.

SECTION I INTRODUCTION AND GENERAL INFORMATION

GENERAL

INTRODUCTION

ENVIRONMENTAL SYSTEMS

FUEL SYSTEM

OIL SYSTEM

LEADING PARTICULARS

ENVIRONMENTAL SYSTEMS

FUEL SYSTEM

OIL SYSTEM

SECTION II ENVIRONMENTAL SYSTEMS

GENERAL

PHYSICAL DESCRIPTION

COOLING SYSTEMS

Equipment Compartment

Payload and Scorer

Downlink Transmitter

HEATING SYSTEMS

Battery Heaters

Pitot Heaters

Slab Heaters

PRESSURIZATION SYSTEMS

Equipment Compartment

Payload Compartment

FUNCTIONAL DESCRIPTION

COOLING SYSTEMS

Equipment Compartment

Payload and Scorer

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SECTION II (Continued)

Downlink Transmitter

HEATING SYSTEMS

Battery Heaters

Pitot Heater

Slab Heaters

PRESSURIZATION SYSTEMS

Equipment Compartment

Payload Compartment

EQUIPMENT REQUIRED

TEST EQUIPMENT

TOOLS

CONSUMABLE MATERIALS

REMOVAL

EQUIPMENT COMPARTMENT COOLING SYSTEMS

Coolant Storage Tank

Cooling Unit

Vent Valve

Shutoff Valve

Cooling Ducts

Heat Exchanger

Wiring Harnesses

Lines, Tubing and Fittings

PAYLOAD AND SCORER COOLING SYSTEM

Coolant Storage Tank

Cooling Unit

Vent Valve

Shutoff Valve

Cooling Ducts

Heat Exchanger

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SECTION II (Continued)

Lines, Tubing and Fittings

Wiring and Harnesses

PRESSURIZATION SYSTEMS

Pressurization Control Valve

Shutoff Valve

Bleed Air Valve

Relief Valve (Pressure/Vacumn)

Vent Valve

Lines, Tubing and Fittings

Wiring and Harnesses

INSTALLATION

EQUIPMENT COMPARTMENT COOLING SYSTEM

Coolant Storage Tank

Cooling Unit

Vent Valve

Shutoff Valve

Cooling Ducts

Lines, Tubing and Fittings

Wiring and Harnesses

PAYLOAD AND SCORER COOLING SYSTEM

Coolant Storage Tank

Cooling Unit

Vent Valve

Shutoff Valve

Cooling Ducts

Heat Exchanger

Lines, Tubing and Fittings

Wiring and Harnesses

PRESSURIZATION SYSTEMS

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ENVIRONMENTAL, FUEL AND OIL SYSTEM MANUAL (Continued)

SECTION II (Continued)

Pressurization Control Valve

Shutoff Valve

Bleed Air Valve

Relief Valve (Pressure/Vacuum)

Vent Valve

Lines, Tubing and Fittings

Wiring and Harnesses

MAINTENANCE

INSPECTION

CLEANING

ADJUSTMENTS

LUBRICATION

OPERATIONAL CHECKOUT

LEAK TESTS

EXTREME CLIMATIC AND ENVIRONMENTAL INFORMATION

SERVICING

COOLING SYSTEMS

HEATING SYSTEMS

PRESSURIZATION SYSTEMS

TROUBLESHOOTING

INSPECTION

SYSTEM ANALYSIS

SECTION III FUEL SYSTEM

GENERAL

PHYSICAL DESCRIPTION

FUEL TANK

IFR Fittings

Filler Cap

Anti-Gravitational Valve

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SECTION II (Continued)

Fuel Tank Sump

Tank Manual Drain Valve

BOOSTER PUMP

EXPULSION (DUMP) VALVE

SHUTOFF VALVE

VENT VALVE

RELIEF VALVE

INERTIA CHECK VALVE

FUEL LEVEL SENSOR

FUEL LINES, TUBING AND FITTINGS

WIRING AND HARNESSES

FUNCTIONAL DESCRIPTION

FUEL TANK

BOOSTER PUMP

EXPULSION (DUMP) VALVE

SHUTOFF VALVE

VENT VALVE

RELIEF VALVE

INERTIA CHECK VALVE

FUEL LEVEL SENSOR

FUEL LINES, TUBING AND FITTINGS

WIRING AND HARNESSES

EQUIPMENT REQUIRED

TEST EQUIPMENT

TOOLS

CONSUMABLE MATERIALS

REMOVAL

FUEL TANK ACCESS DOOR

MANUAL DRAIN VALVE

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ENVIRONMENTAL, FUEL AND OIL SYSTEM MANUAL (Continued)

SECTION III (Continued)

BOOST PUMP

EXPULSION VALVE

SHUTOFF VALVE

VENT VALVE

RELIEF VALVE

INERTIA CHECK VALVE

FUEL LEVEL SENSOR

FUEL LINES, TUBING AND FITTINGS

WIRING AND HARNESSES

INSTALLATION

FUEL TANK ACCESS DOOR

MANUAL DRAIN VALVE

BOOST PUMP

EXPULSION VALVE

SHUTOFF VALVE

VENT VALVE

RELIEF VALVE

INERTIA CHECK VALVE

FUEL LEVEL SENSOR

FUEL LINES, TUBING AND FITTINGS

WIRING AND HARNESSES

MAINTENANCE

INSPECTION

CLEANING

ADJUSTMENTS

LUBRICATION

OPERATIONAL CHECK OUT

LEAK TESTS

EXTREME CLIMATIC AND ENVIRONMENTAL INFORMATION

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ENVIRONMENTAL, FUEL AND OIL SYSTEM MANUAL (Continued)

SECTION III (Continued)

SERVICING

TROUBLESHOOTING

INSPECTION

SYSTEM ANALYSIS

SECTION IV OIL SYSTEM

GENERAL

PHYSICAL DESCRIPTION

STORAGE TANK

Filler fitting

Vent Valve

Drain Plug

OIL LINES, TUBING AND FITTINGS

Inlet Line

Return Line

Vent Line

FUNCTIONAL DESCRIPTION

STORAGE TANK

OIL LINES, TUBING AND FITTINGS

OIL COOLING

EQUIPMENT REQUIRED

TEST EQUIPMENT

TOOLS

CONSUMABLE MATERIALS

REMOVAL

STORAGE TANK

FILLER FITTING

LINES, TUBING AND FITTINGS

INSTALLATION

STORAGE TANK

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SECTION IV (Continued)

FILLER FITTING

LINES, TUBING AND FITTINGS

MAINTENANCE

INSPECTION

CLEANING/FLUSHING

ADJUSTMENTS

LUBRICATION

OPERATIONAL CHECKOUT

LEAK TESTS

EXTREME CLIMATIC AND ENVIRONMENTAL INFORMATION

SERVICEING

TROUBLESHOOTING

INSPECTION

SYSTEM ANALYSIS

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3.3.6 Recovery System Manual.

3.3.6.1 General Content. This manual shall describe the RPV recovery system including the recovery/retrieval sequence. Instructions for component removal and installation, maintenance, servicing, and troubleshooting by organizational (flight line) maintenance personnel shall be provided.

3.3.6.2 Section I, Introduction and Description. This section shall consist of at least three primary paragraph headings; introduction, physical description, and functional description.

3.3.6.2.1 Introduction. The introduction shall identify the components comprising the RPV recovery system and provide leading particulars.

3.3.6.2.2 Physical Description. A complete physical description shall be provided for components comprising the RPV recovery system. The physical description shall include physical characteristics, dimensions, weights, locations, design features, and mounting methods. Operational equipments and range peculiar components shall be included.

3.3.6.2.3 Functional Description. A complete functional description supported by appropriate illustrations/functional block diagrams shall be provided for the recovery system. Recovery system interface with other RPV systems (i.e., electrical, flight control, airframe, etc.)shall be included.

3.3.6.3 Section II, Recovery and Retrieval Sequence. This section shall consist of number of primary paragraphs required for recovery and retrieval sequence of operation. Recovery methods (i.e., mid-air retrieval, surface, etc.), recovery modes (i.e., normal, emergency, automatic), and retrieval methods (i.e., water, land, helicopter, etc) shall be included.

3.3.6.3.1 Recovery Altitudes. Include description of drag parachute deployment, main parachute deployment, deployment limitations (i.e., baroswitch limits, etc.) and identify normal recovery altitudes for mid-air-retrieval.

3.3.6.3.2 Recovery Modes. Include operational sequence and limitations for commanded recovery, emergency recovery, and automatic recovery as applicable.

3.3.6.3.3 Retrieval Sequences. Provide complete description of RPV following mid-air retrieval, surface recovery (land and water) and retrieval by aircraft and watercraft.

3.3.6.4 Section III, Removal and Installation. This section shall consist of at least the following primary paragraph headings; equipment required, parachute installation, parachute removal, component removal and component installation.

3.3.6.4.1 Equipment Required. All equipment and materials required to accomplish installation and removal of parachute system and components shall be identified as required by paragraph 3.3.5.5.3 of this specification.

3.3.6.4.2 Parachute Installation. Procedures for installation of the complete parachute system (i.e., main, drag, stabilization) and containers shall be provided. Post installation tests and inspections shall be included. Explosive safety shall be emphasized.

3.3.6.4.3 Parachute Removal. Procedures for removal of the complete parachute system (i.e., main, drag, stabilization) and containers shall be provided. Explosive safety procedures shall be emphasized.

3.3.6.4.4 Component Removal. Procedures for removal of recovery system components from the RPV shall be provided as required by paragraph 3.2.1.3 of this specification.

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3.3.6.4.5 Component Installation. Procedures for installation of recovery system components on the RPV shall be provided as required by paragraph 3.2.1.3 of this specification.

3.3.6.5 Section IV, Maintenance and Servicing. This section shall consist of at least three primary paragraph headings; equipment required, maintenance and servicing.

3.3.6.5.1 Equipment Required. All equipment and materials required to accomplish maintenance and servicing of the recovery system shall be identified as required by paragraph 3.2.1.2 of this specificaiton.

3.3.6.5.2 Maintenance. Maintenance procedures shall include inspection, cleaning, adjustments, lubrication, operational checkout, and extreme climatic and environmental information as required by paragraph 3.2.1.4 of this specification.

3.3.6.5.3 Servicing. Servicing required incident to maintenance shall be included per paragraph 3.2.1.5 of this specification.

3.3.6.6 Section V, Troubleshooting. This section shall consist of at least two primary paragraph headings; inspection and system analysis.

3.3.6.6.1 Inspection. Procedures for visual inspection of suspected components as an aid to troubleshooting shall be provided as required by paragraph 3.2.1.6 of this specification.

3.3.6.6.2 System Analysis. Troubleshooting procedures for the recovery system shall be tabulated as required by paragraph 3.2.1.6 of this specification.

3.3.6.6.3 Manual Outline. The following is a sample outline of the Recovery System manual. Primary paragraph headings are mandatory for RPV equipped with parachute recovery systems. Paragraph headings shall be arranged as necessary for RPV equipped with other type recovery systems.

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RECOVERY SYSTEM MANUAL

NOTE

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SECTION I INTRODUCTION AND DESCRIPTION

GENEPAL

INTRODUCTION

RECOVERY SYSTEM

LEADING PARTICULARS

PHYSICAL DESCRIPTIONS

DRAG PARACHUTE ASSEMBLY

Drag Parachute

Pilot Parachute

MAIL PARACHUTE ASSEMBLY

Release Mechanism

Load Line

Engagement Parachute

Main Parachute

PARACHUTE CONTAINERS

Drag Parachute Container

Main Parachute Compartment

STABILIZATION PARACHUTE

RISER ASSEMBLIES (GROUND/AIR)

PYROTECHNIC DEVICES

RESISTOR ASSEMBLIES

TIMERS

Range Timers

Lost Carrier (10 - 120 Seconds)

Dual 0 - 5 Seconds

20 seconds

Emergency Chute (5 Seconds)

Command Chute (9 Seconds)

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RECOVERY SYSTEM MANUAL (Continued)

SECTION I (Continued)

BAROMETRIC SWITCHES

Command Chute

Emergency Chute

Adjustable G-Switch Lockout

GRAVITY/IMPACT SWITCH

SALTWATER SWITCH(ES)

LANYARD SWITCH

CONTROL BOXES

Recovery Relay

Relay Control

Squib Control Relay

FUNCTIONAL DESCRIPTION

DRAG PARACHUTE ASSEMBLY

MAIN PARACHUTE ASSEMBLY AND RISER

ELECTRICAL COMPONENTS

Resistors

Timers

Altitude Airspeed Switch

Barometric Switches

Gravity Impact Switch

Saltwater Switch

Lanyard Switch

CONTROL BOXES

Recovery Relay Box

Relay Control Box

Squib Control Relay Box

SECTION II RECOVERY AND RETRIEVAL SEQUENCE

GENERAL

RECOVERY ALTITUDES

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RECOVERY SYSTEM MANUAL (Continued)

SECTION I (Continued)

DRAG DEPLOYMENT

MAIN DEPLOYMENT

RECOVERY MODES

COMMANDED RECOVERY

RECOVERY SEQUENCE

AUTOMATIC RECOVERY

AUTOMATIC RECOVERY SEQUENCE - OPERATIONAL PLUG

AUTOMATIC RECOVERY SEQUENCE - RANGE PLUG

RETRIEVAL SEQUENCES

MARS RETRIEVAL

SURFACE (NON-MARS) RETRIEVAL

SECTION III REMOVAL AND INSTALLATION

GENERAL

EQUIPMENT REQUIRED

TEST EQUIPMENT

TOOLS

CONSUMABLE MATERIALS

PARACHUTE INSTALLATION

PREPARATION

RISER INSTALLATION

ASSEMBLY AND INSTALLATION OF MAIN PARACHUTE RELEASE CHAMBERS

STABILIZATION PARACHUTE INSTALLATION

MARS PARACHUTE INSTALLATION

DRAG PARACHUTE AND CONTAINER INSTALLATION

FINAL COMPONENT INSTALLATION

PARACHUTE REMOVAL

PREPARATION

DRAG PARACHUTE AND CONTAINER REMOVAL

MARS PARACHUTE REMOVAL

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RECOVERY SYSTEM MANUAL (Continued)

SECTION III (Continued)

MAIN PARACHUTE RELEASE CHAMBER REMOVAL

STABILIZATION PARACHUTE REMOVAL

RISER REMOVAL

COMPONENT REMOVAL

PYROTECHNIC DEVICES

RESISTOR ASSEMBLIES

TIMERS

- Range Timers
- Lost Carrier (10 - 120 Seconds)
- Dual 0 - 5 Seconds
- 20 Seconds
- Emergency Chute (5 Seconds)
- Command Chute (9 Seconds)

BAROMETRIC SWITCHES

- Command Chute
- Emergency Chute
- Adjustable G-Switch Lockout

GRAVITY/IMPACT SWITCH(ES)

SALTWATER SWITCH

LANYARD SWITCH

CONTROL BOXES

- Recovery Relay
- Relay Control
- Squib Control Relay

WIRE HARNESSES

COMPONENT INSTALLATION

PYROTECHNIC DEVICES

RESISTOR ASSEMBLIES

TIMERS

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RECOVERY SYSTEM MANUAL (Continued)

SECTION III (continued)

Range Timers

Lost Carrier (10 - 120 Seconds)

Dual 0 - 5 Seconds

20 Seconds

Emergency Chute (5 Seconds)

Command Chute (9 Seconds)

BAROMETRIC SWITCHES

Command Chute

Emergency Chute

Adjustable G-Switch Lockout

GRAVITY/IMPACT SWITCH(ES)

SALTWATER SWITCH

LANYARD SWITCH

CONTROL BOXES

Recovery Relay

Relay Control

Squib Control Relay

WIRE HARNESSES

SECTION IV MAINTENANCE AND SERVICING

GENERAL

EQUIPMENT REQUIRED

TEST EQUIPMENT

TOOLS

CONSUMABLE MATERIALS

MAINTENANCE

INSPECTION

CLEANING

ADJUSTMENTS

LUBRICATION

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RECOVERY SYSTEM MANUAL (Continued)

SECTION IV (Continued)

OPERATIONAL CHECKOUT

EXTREME CLIMATIC AND ENVIRONMENTAL INFORMATION

SERVICING

MAIN PARACHUTE RELEASE CHAMBERS

SECTION V TROUBLESHOOTING

GENERAL

INSPECTION

SYSTEM ANALYSIS

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3.3.7 Electrical Power Supply and Distribution System Manual.

3.3.7.1 General Content. This manual shall describe the RPV electrical power supply and distribution system. Instructions for component removal and installation, maintenance, servicing, and troubleshooting by organizational (flightline) maintenance personnel shall be provided.

3.3.7.2 Section I, Introduction and Description. This section shall consist of at least three primary paragraph headings; introduction, physical description, and functional description.

3.3.7.2.1 Introduction. The introduction shall identify the components comprising the electrical power and distribution system and provide leading particulars.

3.3.7.2.2 Physical Description. A complete physical description shall be provided for components comprising the electrical power supply and distribution system. The physical description shall include physical characteristics, dimensions, weights, locations, design features, and mounting methods. Optional equipments and range components shall be included.

3.3.7.2.3 Functional Description. A complete functional description, supported by functional block diagrams, shall be provided for the electrical power and distribution system. Electrical power and distribution system interface with other RPV systems (i.e., flight control, etc) shall be included. All power sources, control units, and distribution buses shall be included in this description.

3.3.7.3 Section II, Removal and Installation. This section shall consist of at least three primary paragraph headings; equipment required, removal and installation.

3.3.7.3.1 Equipment Required. All equipment and materials required to accomplish removal and installation of components comprising the RPV electrical power and distribution system shall be identified as required by paragraph 3.2.1.2 of this specification.

3.3.7.3.2 Removal. Procedures for removal of electrical system components from the RPV shall be provided as required by paragraph 3.2.1.3 of this specification.

3.3.7.3.3 Installation. Procedures for installation of electrical system components on the RPV shall be provided as required by paragraph 3.2.1.3 of this specification.

3.3.7.4 Section III, Maintenance and Servicing. This section shall consist of at least three primary paragraph headings; equipment required, maintenance and servicing.

3.3.7.4.1 Equipment Required. All equipment and materials required to accomplish maintenance and servicing of the electrical power supply and distribution system shall be identified as required by paragraph 3.2.1.2 of this specification.

3.3.7.4.2 Maintenance. Maintenance procedures shall include inspection, cleaning, adjustments. lubrication, operational checkout, and extreme climatic and environmental information as required by paragraph 3.2.1.4 of this specification.

3.3.7.4.3 Servicing. Servicing required incident to maintenance shall be included per paragraph 3.2.1.5 of this specification.

3.3.7.5 Section IV, Troubleshooting. This section shall consist of at least three primary paragraph headings; inspection, system analysis, AC/DC loads analysis.

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3.3.7.5.1 Inspection. Procedures for visual inspection of suspected components as an aid to troubleshooting shall be provided as required by paragraph 3.2.1.6 of this specification.

3.3.7.5.2 System Analysis. Troubleshooting procedures for the electrical system shall be tabulated as required by paragraph 3.2.1.6 of this specification.

3.3.7.5.3 AC/DC Loads Analysis. AC and DC electrical loads analysis shall be tabulated using the format in figures 6,7, and 8.

3.3.7.6 Manual Outline. The following is a sample outline of the Electrical Power Supply and Distribution System manual. Primary paragraph headings are mandatory. Subordinate paragraph headings are representative and will be dependent upon equipment comprising the electrical system of the RPV being covered by the manuals.

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ELECTRICAL POWER SUPPLY AND DISTRIBUTION MANUAL

NOTE

Front matter shall be arranged as required by paragraph 3.1.8 of this specification.

SECTION I INTRODUCTION AND DESCRIPTION

GENERAL

INTRODUCTION

POWER SOURCES

CONTROL AND DISTRIBUTION

LEADING PARTICULARS

PHYSICAL DESCRIPTION

POWER SOURCE COMPONENTS

Starter-Generator

Voltage Regulator

Inverter

Batteries

POWER DISTRIBUTION AND CONTROL COMPONENTS

Engine Start Power Receptacle

Main Control Umbilical Distribution Box

Main Power Control/Distribution Box

Special Devices Interconnect Box

Auxiliary Power Control Box

Relays

Switches

Wiring and Harnesses

FUNCTIONAL DESCRIPTION

POWER SOURCE COMPONENTS

Starter-Generator

Voltage Regulator

Inverter

Batteries

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ELECTRICAL POWER SUPPLY AND DISTRIBUTION MANUAL (Continued)

SECTION I (Continued)

POWER DISTRIBUTION AND CONTROL COMPONENTS

Engine Start Power Receptacle

Main Control Umbilical Distribution Box

Main Power Control/Distribution Box

Special Devices Interconnect Box

Auxiliary Power Control Box

Relays

Switches

Wiring Harnesses

POWER DISTRIBUTION BUSES

Generator and Battery Buses

Main Bus

ECM Bus

Autopilot Bus

Special Devices Bus

SECTION II REMOVAL AND INSTALLATION

GENERAL

EQUIPMENT REQUIRED

TEST EQUIPMENT

TOOLS

REMOVAL

STARTER-GENERATOR

VOLTAGE REGULATOR

VOLTAGE REGULATOR MOUNTING BASE

INVERTER

BATTERY

ENGINE START POWER RECEPTACLE

MAIN UMBILICAL CONTROL DISTRIBUTION BOX

MAIN POWER CONTROL/DISTRIBUTION BOX

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SECTION II (Continued)

SPECIAL DEVICES INTERCONNECT BOX

AUXILIARY POWER CONTROL BOX

SWITCHES

WIRING HARNESSES

INSTALLATION

STARTER-GENERATOR

VOLTAGE REGULATOR

VOLTAGE REGULATOR MOUNTING BASE

INVERTER

BATTERY

ENGINE START POWER RECEPTACLE

MAIN CONTROL UMBILICAL DISTRIBUTION BOX

MAIN POWER CONTROL/DISTRIBUTION BOX

SPECIAL DEVICES INTERCONNECT BOX

AUXILIARY POWER CONTROL BOX

SWITCHES

WIRING HARNESSES

SECTION III MAINTENANCE AND SERVICING

GENERAL

EQUIPMENT REQUIRED

TEST EQUIPMENT

TOOLS

CONSUMABLE MATERIALS

MAINTENANCE

INSPECTION

CLEANING

ADJUSTMENTS

LUBRICATION

OPERATIONAL CHECKOUT

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ELECTRICAL POWER SUPPLY AND DISTRIBUTION MANUAL (Continued)

SECTION III (Continued)

EXTREME CLIMATIC AND ENVIRONMENTAL INFORMATION

SERVICING

SECTION IV TROUBLESHOOTING

GENERAL

INSPECTION

SYSTEM ANALYSIS

AC/DC LOADS ANALYSIS

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3.3.8 Electronics and Guidance Systems Manual.

3.3.8.1 General Content. This manual shall describe the RPV electronics and guidance systems. Procedures for removal and installation of electronics and guidance systems components, maintenance, servicing, and troubleshooting by organizational (flightline) maintenance personnel shall be provided.

3.3.8.2 Section I, Introduction and Description. This section shall consist of at least three primary paragraph headings; introduction, physical description, and functional description.

3.3.8.2.1 Introduction. The introduction shall identify the major components comprising the RPV electronics and guidance systems and provide leading particulars.

3.3.8.2.2 Physical Description. A complete physical description shall be provided for components comprising the electronics and guidance systems. The physical description shall include physical characteristics, dimensions, weights, locations, design features, and mounting methods.

3.3.8.2.3 Functional Description. A complete functional description, supported by functional block diagrams and/or data flow diagrams, shall be provided for the electronics and guidance systems. The functional description shall be arranged in the most logical sequence possible to best describe the total system level operation. Electronics and guidance systems interface with other RPV systems (i.e., electrical, flight control, etc.) shall be included.

3.3.8.3 Section II, Removal and Installation. This section shall consist of at least three primary paragraph headings; equipment require, removal and installation.

3.3.8.3.1 Equipment Required. All equipment and materials required to accomplish removal and installation of electronics and guidance systems components shall be identified as required by paragraph 3.2.1.2 of this specification.

3.3.8.3.2 Removal. Procedures for removal of electronics and guidance systems components from the RPV shall be provided as required by paragraph 3.2.1.3 of this specification.

3.3.8.3.3 Installation. Procedures for installation of electronics and guidance systems components on the RPV shall be provided as required by paragraph 3.2.1.3 of this specification.

3.3.8.4 Section III, Maintenance and Servicing. This section shall consist of at least three primary paragraph headings; equipment required, maintenance and servicing.

3.3.8.4.1 Equipment Required. All equipment and materials required to accomplish maintenance and servicing of electronics and guidance systems components shall be identified as required by paragraph 3.2.1.2 of this specification.

3.3.8.4.2 Maintenance. Maintenance procedures shall include inspection, cleaning, adjustments, lubrication, operational checkout, and extreme climatic and environmental information as required by paragraph 3.2.1.4 of this specification.

3.3.8.4.3 Servicing. Servicing required incident to maintenance shall be provided as required by paragraph 3.2.1.5 of this specification.

3.3.8.5 Section IV, Troubleshooting. This section shall consist of at least two primary paragraph headings; inspection and systems analysis.

3.3.8.5.1 Inspection. Procedures for accomplishing visual inspections of

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components as a part of troubleshooting shall be provided as required by paragraph 3.2.1.6 of this specification.

3.3.8.5.2 Systems Analysis. Troubleshooting procedures shall be tabulated as required by paragraph 3.2.1.6 of this specification.

3.3.8.6 Manual Outline. The following is a sample outline of the Electronics and Guidance Systems manual. Primary sideheads are mandatory. Subordinate sideheads listed under maintenance are mandatory. Subordinate sideheads in section I and II are representative and actual headings shall be dependent upon equipment comprising the electronics and guidance systems of the specific RPV covered by the manual.

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ELECTRONICS AND GUIDANCE SYSTEMS MANUAL

NOTE

Front matter shall be arranged as required by paragraph 3.1.8 of this specification.

SECTION I INTRODUCTION AND DESCRIPTION

GENERAL

INTRODUCTION

GUIDANCE SYSTEM

LEADING PARTICULARS

PHYSICAL DESCRIPTION

LORAN RECEIVER SUBSYSTEM

BULK STORAGE SUBSYSTEM

DOPPLER RADAR SUBSYSTEM

RADAR ALTIMETER SUBSYSTEM

DIGITAL PROCESSOR SUBSYSTEM

ATTITUDE AND HEADING REFERENCE SUBSYSTEM

MICROWAVE COMMAND GUIDANCE SYSTEM

INTERFACE AND CONTROL SUBSYSTEM

FUNCTIONAL DESCRIPTION

LORAN RECEIVER SUBSYSTEM

BULK STORAGE SUBSYSTEM

DOPPLER RADAR SUBSYSTEM

RADAR ALTIMETER SUBSYSTEM

DIGITAL PROCESSOR SUBSYSTEM

ATTITUDE AND HEADING REFERENCE SUBSYSTEM

MICROWAVE COMMAND GUIDANCE SYSTEM

INTERFACE AND CONTROL SUBSYSTEM

SECTION II REMOVAL AND INSTALLATION

GENERAL

EQUIPMENT REQUIRED

TEST EQUIPMENT

TOOLS

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ELECTRONICS AND GUIDANCE SYSTEMS MANUAL (Continued)

SECTION II (Continued)

CONSUMABLE MATERIALS

REMOVAL

LORAN RECEIVER

BULK STORAGE UNIT

DOPPLER RADAR

RADAR ALTIMETER

DIGITAL PROCESSOR

ATTITUDE AND HEADING REFERENCE UNIT

MCGS TRANSPONDER

INTERFACE AND CONTROL UNIT

INSTALLATION

LORAN RECEIVER

BULK STORAGE UNIT

DOPPLER RADAR

RADAR ALTIMETER

DIGITAL PROCESSOR

ATTITUDE AND HEADING REFERENCE UNIT

MCGS TRANSPONDER

INTERFACE AND CONTROL UNIT

SECTION III MAINTENANCE AND SERVICING

GENERAL

EQUIPMENT REQUIRED

TEST EQUIPMENT

TOOLS

CONSUMABLE MATERIALS

MAINTENANCE

INSPECTION

CLEANING

ADJUSTMENTS

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ELECTRONICS AND GUIDANCE SYSTEMS MANUAL (Continued)

SECTION III MAINTENANCE AND SERVICING

LUBRICATION

OPERATIONAL CHECKOUT

EXTREME CLIMATIC AND ENVIRONMENTAL INFORMATION

SERVICING

SECTION IV TROUBLESHOOTING

GENERAL

INSPECTION

SYSTEMS ANALYSIS

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3.3.9 Flight Control System Manual.

3.3.9.1 General Content. This manual shall describe the RPV flight control system. Procedures for removal and installation of flight control system components, maintenance, servicing, and troubleshooting by organizational (flightline) maintenance personnel shall be provided.

3.3.9.2 Section I, Introduction and Description. This section shall consist of at least three primary paragraph headings; introduction, physical description and functional description.

3.3.9.2.1 Introduction. The introduction shall identify the components comprising the RPV flight control system and provide leading particulars.

3.3.9.2.2 Physical Description. A complete physical description shall be provided for components comprising the flight control system. The physical description shall include physical characteristics, dimensions, weights, locations, design features and mounting methods.

3.3.9.2.3 Functional Description. A complete functional description, supported by functional block diagrams, shall be provided for the flight control system. The functional description shall be arranged in the most logical sequence possible to best describe the total system level operation. Flight control system interface with other RPV systems (i.e., guidance, electrical, etc) shall be included.

3.3.9.3 Section II, Removal and Installation. This section shall consist of at least three primary paragraph headings; equipment required, removal and installation.

3.3.9.3.1 Equipment Required. All equipment and materials required to accomplish removal and installation of flight control system components shall be identified as required by paragraph 3.2.1.2 of this specification.

3.3.9.3.2 Removal. Procedures for removal of flight control system components from the RPV shall be provided as required by paragraph 3.2.1.3 of this specification.

3.3.9.3.3 Installation. Procedures for installation of flight control system components on the RPV shall be provided as required by paragraph 3.2.1.3 of this specification.

3.3.9.4 Section III, Maintenance and Servicing. This section shall consist of at least three primary paragraph headings; equipment required, maintenance and servicing.

3.3.9.4.1 Equipment Required. All equipment and materials required to accomplish maintenance and servicing of flight control system components shall be identified as required by paragraph 3.2.1.2 of this specification.

3.3.9.4.2 Maintenance. Maintenance procedures shall include inspection, cleaning, adjustments, lubrication, operational checkout and extreme climatic and environmental information as required by paragraph 3.2.1.4 of this specification.

3.3.9.4.3 Servicing. Servicing required incident to maintenance shall be provided was required by paragraph 3.2.1.5 of this specification.

3.3.9.5 Section IV, Troubleshooting. This section shall consist of at least two primary paragraph headings; inspection and systems analysis.

3.3.9.5.1 Inspection. Procedures for accomplishing visual inspections of components as a part of troubleshooting shall be provided as required by paragraph 3.2.1.6 of this specification.

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3.3.9.5.2 Systems Analysis. Troubleshooting procedures shall be tabulated as required by paragraph 3.2.1.6 of this specification.

3.3.9.6 Manual Outline. The following is a sample outline of the Flight Control System manual. Primary sideheads are mandatory. Subordinate sideheads listed under maintenance are mandatory. Subordinate sideheads in sections I and II are representative and actual headings shall be dependent upon equipment comprising the flight control system of the specific RPV covered by the manual.

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FLIGHT CONTROL SYSTEMS MANUAL

NOTE

Front matter shall be arranged as required by paragraph 3.1.8 of this specification.

SECTION I INTRODUCTION AND DESCRIPTION

GENERAL

INTRODUCTION

FLIGHT CONTROL SYSTEM

LEADING PARTICULARS

PHYSICAL DESCRIPTION

FLIGHT CONTROL BOX

FLIGHT CONTROL BOX RACK

LATERAL CONTROLLER UNIT

PROGRAMMER

PROGRAMMER RACK

FLIGHT HEADING/REFERENCE SYSTEM

PITOT-STATIC SYSTEM

AILERON ACTUATOR

ELEVATOR ACTUATOR

RUDDER ACTUATOR

RUDDER ACTUATOR FILTER

FUNCTIONAL DESCRIPTION

FLIGHT CONTROL SYSTEM

FLIGHT CONTROL BOX

LATERAL CONTROLLER UNIT

PROGRAMMER

FLIGHT CONTROL COMMANDS AND MODES

COMPASS SYSTEM

FLIGHT HEADING/REFERENCE SYSTEM

ALIERON ACTUATOR

ELEVATOR ACTUATOR

RUDDER ACTUATOR

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SECTION I (Continued)

RUDDER ACTUATOR FILTER

SECTION II REMOVAL AND INSTALLATION

GENERAL

EQUIPMENT REQUIRED

TEST EQUIPMENT

TOOLS

CONSUMABLE MATERIALS

REMOVAL

FLIGHT CONTROL BOX

FLIGHT CONTROL BOX RACK

LATERAL CONTROLLER UNIT

PROGRAMMER

PROGRAMMER RACK

GYRO AND SYNCHRONIZER ASSEMBLY

FLUX VALVE

PITOT-STATIC SYSTEM

Pitot-Static Tube

Lines and Tubing

AILERON ACTUATOR

ELEVATOR ACTUATOR

RUDDER ACTUATOR

RUDDER ACTUATOR FILTER

WIRING HARNESSES

INSTALLATION

FLIGHT CONTROL BOX

FLIGHT CONTROL BOX RACK

LATERAL CONTROLLER UNIT

PROGRAMMER

PROGRAMMER RACK

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FLIGHT CONTROL SYSTEMS MANUAL (Continued)

SECTION II (continued)

GYRO AND SYNCHRONIZER ASSEMBLY

FLUX VALVE

PITOT-STATIC SYSTEM

Pitot-Static Tube

Lines and Tubing

AILERON ACTUATOR

ELEVATOR ACTUATOR

RUDDER ACTUATOR

RUDDER ACTUATOR FILTER

WIRING HARNESSES

SECTION III MAINTENANCE AND SERVICING

GENERAL

EQUIPMENT REQUIRED

TEST EQUIPMENT

TOOLS

CONSUMABLE MATERIALS

MAINTENANCE

INSPECTION

CLEANING

ADJUSTMENTS

LUBRICATION

OPERATIONAL CHECKOUT

EXTREME CLIMATIC AND ENVIRONMENTAL INFORMATION

SERVICING

SECTION IV TROUBLESHOOTING

GENERAL

INSPECTION

SYSTEMS ANALYSIS

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3.3.10 Utility Systems Manual.

3.3.10.1 General Content. This manual shall describe and provide component removal and installation, maintenance, servicing and troubleshooting instructions for use by organizational (flightline) maintenance personnel for miscellaneous systems/equipment which do not function in one of the systems covered by one of the other manuals required by this specification. Examples of equipment included in this manual are; tracking beacons, range beacons, air inlet (antiwindmill) covers and auxiliary equipments.

3.3.10.1.2 Payload Systems. The procuring activity may elect to include the reconnaissance systems, armament systems, scorer systems, etc in the utility systems manual in lieu of obtaining a separate payload and reconnaissance system manual. If this option is elected, each payload system will be covered in a separate section added to the utility systems manual. The content of each added section shall be arranged in same format and depth as shown in manual outline for the Payload and Reconnaissance System manual (paragraph 3.3.15 of this specification).

3.3.10.2 Section I, Introduction and General Information. This section shall consist of at least two primary paragraph headings; introduction and leading particulars.

3.3.10.2.1 Introduction. The introduction shall identify the systems covered in the manual and tabulate the components comprising each system.

3.3.10.2.2 Leading Particulars. Leading particulars for each system/equipment covered by the manual shall be tabulated.

3.3.10.3 Section II, Beacons. This section shall describe and provide removal, installation, maintenance, servicing, and troubleshooting instructions for range beacons, tracking beacons, post recovery beacons, etc. used on the RPV covered by the manual. This section shall consist of the following primary paragraph headings; physical description, functional description, equipment required, removal, installation, maintenance, servicing and troubleshooting.

3.3.10.3.1 Physical Description. A complete physical description shall be provided for each beacon covered in this section. The physical description shall include physical characteristics, dimensions, weights, locations, design features and mounting methods.

3.3.10.3.2 Functional Description. A complete functional description shall be provided for each beacon included in this section.

3.3.10.3.3 Equipment Required. All equipment and materials required to accomplish removal, installation, maintenance, servicing, and troubleshooting shall be identified as required by paragraph 3.2.1.3 of this specification.

3.3.10.3.4 Removal. Procedures for removal of each beacon and associated components (i.e., antenna, cables, etc) shall be provided as required by paragraph 3.2.1.3 of this specification.

3.3.10.3.5 Installation. Procedures for installation of each beacon and associated components (i.e., antenna, cables, etc) shall be provided as required by paragraph 3.2.1.3 of this specification.

3.3.10.3.6 Maintenance. Maintenance procedures shall include inspection, cleaning, adjustments, lubrication, operational checkout and extreme climatic and environmental information as required by paragraph 3.2.1.4 of this specification.

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3.3.10.3.7 Servicing. Servicing required incident to maintenance shall be provided as required by paragraph 3.2.1.5 of this specification.

3.3.10.3.8 Troubleshooting. Troubleshooting procedures shall be provided for each beacon included in this section as required by paragraph 3.2.1.6 of this specification.

3.3.10.4 Section III and Subsequent Sections. The number of sections included in the utility systems manual is dependent upon the miscellaneous systems/equipment utilized on the specific RPV being covered by the manual. Each system/equipment included shall be covered in a separate section. Each added section shall contain at least the same primary paragraph headings and depth of coverage as described for section II in paragraphs 3.3.10.3 through 3.3.10.3.8 above.

3.3.10.5 Manual Outline. The following is a sample outline for the Utility Systems manual. Primary sideheads are mandatory for sections shown in the outline and any subsequent sections added. Subordinate sideheads listed under maintenance and troubleshooting are mandatory. Section titles and subordinate sideheads identifying equipment are representative. Headings will depend on systems/equipment covered by the manual.

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UTILITY SYSTEMS MANUAL

NOTE

Front matter shall be arranged as required by paragraph 3.1.8 of this specification

SECTION I INTRODUCTION AND GENERAL INFORMATION

GENERAL

INTRODUCTION

BEACONS

AIR INLET COVERS

LEADING PARTICULARS

BEACONS

AIR INLET COVERS

SECTION II BEACONS

GENERAL

PHYSICAL DESCRIPTION

RANGE BEACONS/ANTENNA/CABLES

TRACKING BEACONS/ANTENNA/CABLES

POST RECOVERY BEACONS/ANTENNA/CABLES

FUNCTIONAL DESCRIPTION

RANGE BEACONS/ANTENNA/CABLES

TRACKING BEACONS/ANTENNA/CABLES

POST RECOVERY BEACONS/ANTENNA/CABLES

EQUIPMENT REQUIRED

TEST EQUIPMENT

TOOLS

CONSUMABLE MATERIALS

REMOVAL

RANGE BEACONS/ANTENNA/CABLES

TRACKING BEACONS/ANTENNA/CABLES

POST RECOVERY BEACONS/ANTENNA/CABLES

INSTALLATION

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UTILITY SYSTEMS MANUAL (Continued)

SECTION II (Continued)

RANGE BEACONS/ANTENNA/CABLES

TRACKING BEACONS/ANTENNA/CABLES

POST RECOVERY BEACONS/ANTENNA/CABLES

MAINTENANCE

INSPECTION

CLEANING

ADJUSTMENTS

LUBRICATION

OPERATIONAL CHECKOUT

EXTREME CLIMATIC AND ENVIRONMENTAL INFORMATION

TROUBLESHOOTING

INSPECTION

SYSTEMS ANALYSIS

SECTION III AIR INLET COVERS

GENERAL

PHYSICAL DESCRIPTION

AIR INLET COVERS

RELEASE CHAMBERS

FUNCTIONAL DESCRIPTION

AIR INLET COVERS

RELEASE CHAMBERS

EQUIPMENT REQUIRED

TEST EQUIPMENT

TOOLS

CONSUMABLE MATERIALS

REMOVAL

AIR INLET COVERS

RELEASE CHAMBERS

INSTALLATION

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UTILITY SYSTEMS MANUAL (Continued)

SECTION III (Continued)

AIR INLET COVERS

RELEASE CHAMBERS

MAINTENANCE

INSPECTION

CLEANING

ADJUSTMENTS

LUBRICATION

OPERATIONAL CHECKOUT

EXTREME CLIMATIC AND ENVIRONMENTAL INFORMATION

TROUBLESHOOTING

INSPECTION

SYSTEMS ANALYSIS

NOTE

Subsequent sections may be added to cover additional utility systems/equipment contained in specific RPV being covered and payload systems as reflected in paragraph 3.3.10.1.2 of this specification. Each added section shall be arranged as shown for sections II and III above. Primary sideheads shall be the same in each added section.

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3.3.11 Decontamination Manual.

3.3.11.1 General Content. This manual shall identify the preventative maintenance procedures required to accomplish complete decontamination of the RPV following surface recovery. Procedures shall include preliminary decontamination (normally accomplished at the recovery site) and detailed decontamination (normally accomplished at the support base).

3.3.11.2 Section I, Introduction and General Information. This section shall consist of at least two primary paragraph headings; introduction and decontamination.

3.3.11.2.1 Introduction. The introduction shall identify the symptoms of corrosion (saltwater, oxidation, etc.) and possible impact damage and operational degradation resulting from incomplete or lack of effective decontamination. Exposure limits (saltwater, extreme humidity, etc) shall be included.

3.3.11.2.2 Decontamination. The types of decontamination shall be identified and purpose of each stated. Decontamination methods(i.e., cleaning, application of corrosion preventative materials, drying methods, etc) shall be identified. Procedures for decontamination of radioactive materials shall not be included but standard AF procedures shall be referenced.

3.3.11.3 Section II, Preparation. This section shall consist of at least four primary paragraph headings; equipment required, retrieval, safety (uncock), preparation for decontamination.

3.3.11.3.1 Equipment Required. All equipment and materials required to accomplish retrieval, safety (uncock), preliminary and detailed decontamination shall be identified as required by paragraph 3.2.1.2 of this specification.

3.3.11.3.2 Retrieval. Procedures shall be provided for RPV retrieval by hellicopter ground vehicle and watercraft. Methods of transportation of RPV to support base shall be included.

3.3.11.3.3 Safety (Uncock). Procedures for safing the RPV electrical systems, explosive devices, hazardous liquids (i.e., ammonia, etc.) shall be provided. Safing procedures shall be arranged in the sequence that will present the least hazard to maintenance personnel.

3.3.11.3.4 Preparation for Decontamination. Procedures for preparing the RPV for decontamination shall be provided. This includes removal of access panels, doors, drain plugs, and protective covers. Inspections to determine extent of decontamination and disassembly required shall be provided. Procedures for reporting damage beyond the capability of organizational (flightline) maintenance shall be included.

3.3.11.4 Section III, Preliminary Decontamination. This section shall include procedures for accomplishing preliminary decontamination at the recovery site to prevent systems degradation during transportation of RPV to support base. Primary and subordinate paragraph headings shall be dependent upon systems/equipment comprising the specific RPV being covered by the manual.

3.3.11.5 Section IV, Detailed Decontamination. This section shall consist of at least five primary paragraph headings; inspection, disassembly, decontamination, assembly, and operational checkout. Subordinate paragraph headings shall be dependent upon the systems/equipment comprising the RPV being covered by the manual.

3.3.11.5.1 Inspection. Procedures for inspecting the RPV to determine extent of disassembly and decontamination required shall be provided. Detection and documentation requirements for damage incurred during water recovery shall be included.

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3.3.11.5.2 Disassembly. Procedures for disassembly of the RPV for decontamination shall be provided. Reference to procedures in other manuals required by this specification will suffice.

3.3.11.5.3 Decontamination. Complete procedures for removal of contaminated residue, drying components, and determining serviceability of systems/equipment shall be provided. Equipment and related procedures may be tabulated.

3.3.11.5.4 Assembly. Procedures for assembly of the RPV following decontamination shall be provided. Reference to procedures in other manuals required by this specification will suffice.

3.3.11.5.5 Operational Checkout. Operational checkout requirements following decontamination shall be identified. Procedures contained in other manuals shall be referenced. Procedures not provided in other manuals shall be provided in this paragraph.

3.3.11.6 Manual Outline. The following is a sample outline of the Decontamination manual. Primary and subordinate sideheads shall be included as required by paragraphs 3.3.11.2 through 3.3.11.5.5 above.

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DECONTAMINATION MANUAL

NOTE

Front matter shall be arranged as required by paragraph 3.1.8 of this specification.

SECTION I INTRODUCTION AND GENERAL INFORMATION

GENERAL

INTRODUCTION

CORROSION

LONG TERM DAMAGE

DECONTAMINATION

PURPOSE

METHODS

SECTION II PREPARATION

GENERAL

EQUIPMENT REQUIRED

HANDLING AND SERVICE EQUIPMENT

TEST EQUIPMENT

TOOLS

CONSUMABLE MATERIALS

RETRIEVAL

BOAT

HELICOPTER

GROUND VEHICLE

SAFETY (UNCOCK)

EXPLOSIVES

BATTERY

COOLANT (AMMONIA)

PREPARATION FOR DECONTAMINATION

ACCESS PANELS AND DOORS

DRAIN PLUGS

INSPECTION

DEPOT RECONDITIONING

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DECONTAMINATION MANUAL (Continued)

SECTION II (Continued)

DISASSEMBLY

SECTION III PRELIMINARY DECONTAMINATION

GENERAL

SCORER COMPARTMENT

SCORER MATERIAL

AIRBRAME

FUSELAGE AND EMPENNAGE

EQUIPMENT COMPARTMENTS

WINGS AND PODS

POWER PLANT

ENGINE

ACCESSORIES

PARACHUTES

SECTION IV DETAILED DECONTAMINATION

GENERAL

DISASSEMBLY

INSPECTION

DECONTAMINATION

ASSEMBLY

OPERATIONAL CHECKOUT

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3.3.12 Upload, Pre/Postflight and Download Manual.

3.3.12.1 General Content. This manual shall provide procedures to prepare the RPV for upload on the launch platform, upload, post upload testing and preflight requirements. Procedures for postflight, safety of RPV, and download shall be provided. Maintenance auhtorized while the RPV is uploaded shall be identified and maintenance procedures provided or referenced if contained in another manual required by this specification. This manual shall be prepared to cover all authorized launch platforms (i.e., aircraft, ground launcher) and all authorized configurations of each launch platform.

3.3.12.2 Section I, Introduction and General Information. This section shall consist of at least two primary paragraph headings; introduction, inspection and checkout.

3.3.12.2.1 Introduction. The introduction shall identify and briefly describe the launch platforms authorized. A brief description of upload/download sequence of events shall be included.

3.3.12.2.2 Inspection and Checkout. The pre-upload, mission requirements, post-upload inspection procedures and checks shall be identified in normal sequence of accomplishment. Various flight configurations and related requirements shall be identified. Reference to other manuals applicable to the procedures contained in this manual shall be included and applicable procedure/test identified.

3.3.12.3 Section II, Upload. This section shall consist of at least three primary paragraph headings; equipment required, assembly and servicing, and upload. Additional primary paragraph headings will be dependent upon number of launch platforms authorized and systems/equipment (i.e., fuel pods, chaff pods, etc) applicable to specific RPV covered by the manual.

3.3.12.3.1 Equipment Required. All equipment and materials required to accomplish RPV preparation and upload shall be identified as required by paragraph 3.2.1.2 of this specification.

3.3.12.3.2 Assembly and Servicing. A brief listing of actions required to verify the RPV is fully assembled, mission requirements accomplished, serviced and preparations for upload completed shall be provided.

3.3.12.3.3 Upload. Procedures for transporting the RPV to upload area, preparation of launch platform, and RPV upload shall be provided. Safety requirements shall be emphasized throughout the procedure. Upload on each authorized launch platform (i.e., aircraft, ground launcher) shall be covered under separate primary paragraph headings.

3.3.12.4 Section III, Pre/Postflight. This section shall consist of at least five primary paragraph headings; equipment required, post upload checks, preflight test, preflight inspections, postflight inspections. Post upload checks and preflight tests for each authorized launch platform (i.e., aircraft, ground launcher) shall be provided under separate primary paragraph headings.

3.3.12.4.1 Equipment Required. All equipment and materials required to accomplish pre/postflight tasks of the RPV shall be identified as required by paragraph 3.2.1.2 of this specification.

3.3.12.4.2 Post-Upload Checks. All operational checks (i.e., direct control checks, remote command checks, etc) required to verify system operational after load shall be included for each authorized platform.

3.3.12.4.3 Preflight Tests. All preflight tests (i.e., navigation systems, beacons, recovery systems, etc) shall be included.

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3.3.12.4.4 Preflight Inspections. All required preflight inspections shall be provided. Special procedures (if applicable) for RPV in alert status shall be provided.

3.3.12.4.5 Postflight Inspections. All required postflight (launch/captive) inspection procedures shall be provided. Applicable inspection work cards shall be referenced.

3.3.12.5 Section IV, Download. This section shall consist of at least two primary paragraph headings; equipment required and download. Additional primary paragraph headings shall be included for each authorized launch platform (i.e., aircraft, ground launcher) and for fuel/chaff pods if applicable to RPV being covered by the manual.

3.3.12.5.1 Equipment Required. All equipment and materials required to accomplish download tasks shall be identified as required by paragraph 3.2.1.2 of this specification.

3.3.12.5.2 Download. Complete procedures for safety of RPV systems and download of RPV, pods, etc, shall be provided for each authorized launch platform.

3.3.12.6 Section V, Maintenance. This section shall consist of at least two primary paragraph headings; equipment required and maintenance. Procedures shall be included for each authorized launch platform.

3.3.12.6.1 Equipment Required. All equipment and tools required for accomplishment of maintenance procedures shall be identified as required by paragraph 3.2.1.2 of this specification.

3.3.12.6.2 Maintenance. Maintenance procedures authorized for RPV uploaded on the launch platform shall be identified and procedures shall be provided or referenced as applicable. Checkout procedures required after maintenance shall be provided or referenced as applicable.

3.3.12.7 Section VI, Post Recovery. This section shall consist of at least two primary paragraph headings; equipment required and post recovery uncock. The procedures in this section shall prepare the RPV for safe transportation from the recovery area to the maintenance facility.

3.3.12.7.1 Equipment Required. All equipment and materials required to accomplish post recovery tasks shall be identified as required by paragraph 3.2.1.2 of this specification.

3.3.12.7.2 Post Recovery Uncock. Procedures for safing the RPV, installation of protective covers, and preparation for transporting the RPV to the maintenance facility shall be provided. Safety shall be emphasized throughout the procedure.

3.3.12.8 Manual Outline. The following is a sample outline of the Upload, Pre/Postflight and Download manual. Primary and subordinate paragraphs shall be included as required by paragraphs 3.3.12.2 through 3.3.12.7.2 above.

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UPLOAD, PRE/POSTFLIGHT AND DOWNLOAD MANUAL

NOTE

Front matter shall be arranged as required by paragraph 3.1.8 of this specification.

SECTION I INTRODUCTION AND GENERAL INFORMATION

GENERAL

INTRODUCTION

AIR LAUNCH

GROUND LAUNCH

UPLOAD AND DOWNLOAD

INSPECTION AND CHECKOUT

PRE-UPLOAD CHECKS

POST UPLOAD CHECKS

PREFLIGHT INSPECTION

POSTFLIGHT INSPECTION

POST UPLOAD MAINTENANCE

POST RECOVERY

SECTION II UPLOAD

GENERAL

EQUIPMENT REQUIRED

TEST EQUIPMENT

HANDLING EQUIPMENT

TOOLS

CONSUMABLE MATERIALS

RPV ASSEMBLY AND SERVICING

ASSEMBLY

SERVICING

WEIGHT AND BALANCE

UPLOAD - AIR LAUNCH

PREPARATION

Launch Aircraft

RPV

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SECTION II (Continued)

UPLOAD PROCEDURE (DC-130A)

UPLOAD PROCEDURE (DC-130E)

UPLOAD - GROUND LAUNCH

PREPARATION

Launcher

RPV

ROCKET MOTOR INSTALLATION/ALIGNMENT

UPLOAD PROCEDURE (GROUND LAUNCHER)

UPLOAD EXTERNAL PODS (FUEL/STORES)

PREPARATION

PROCEDURE

SECTION III PRE/POSTFLIGHT

GENERAL

EQUIPMENT REQUIRED

TEST EQUIPMENT

TOOLS

CONSUMABLE MATERIALS

POST UPLOAD CHECKS (RPV ON LAUNCH AIRCRAFT)

DIRECT CONTROL CHECKS

Preparation

Procedure

REMOTE COMMAND RESPONSE CHECKS

Preparation

Procedure

WEAPON INTERFACE

Preparation

Procedure

POST UPLOAD CHECKS (RPV ON GROUND LAUNCHER)

DIRECT CONTROL CHECKS

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UPLOAD, PRE/POSTFLIGHT AND DOWNLOAD MANUAL (Continued)

SECTION III (Continued)

Preparation

Procedure

REMOTE COMMAND RESPONSE CHECKS

Preparation

Procedure

WEAPON INTERFACE

Preparation

Procedure

PREFLIGHT TESTS

NAVIGATION SYSTEM

DOPPLER

BEACONS

RECOVERY SYSTEM (CONFIDENCE TEST)

PREFLIGHT INSPECTIONS

STANDBY (ALERT) STATUS

POST FLIGHT INSPECTIONS

CAPTIVE

AFTER LAUNCH

SECTION IV DOWNLOAD

GENERAL

EQUIPMENT REQUIRED

TEST EQUIPMENT

HANDLING EQUIPMENT

TOOLS

CONSUMABLE MATERIALS

DOWNLOAD FROM LAUNCH AIRCRAFT

SAFETY RPV SYSTEMS (UNCOCK)

WEAPON DOWNLOAD

PREPARATION FOR DOWNLOAD

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SECTION III (Continued)

DOWNLOAD PROCEDURE (DC-130A)

DOWNLOAD PROCEDURE (DC-130E)

DOWNLOAD FROM GROUND LAUNCHER

SAFETY RPV SYSTEMS (UNCOCK)

WEAPON DOWNLOAD

PREPARATION FOR DOWNLOAD

DOWNLOAD PROCEDURE

DOWNLOAD EXTERNAL PODS (FUEL/STORES)

PREPARATION

PROCEDURE

SECTION V MAINTENANCE

GENERAL

EQUIPMENT REQUIRED

TEST EQUIPMENT

TOOLS

CONSUMABLE MATERIALS

UNDER WING MAINTENANCE (RPV UPLOADED - LAUNCH AIRCRAFT)

PREPARATION

UNDERWING MAINTENANCE REQUIREMENTS

POST MAINTENANCE REQUIREMENTS AND CHECKOUT

EXTREME CLIMATIC AND ENVIRONMENTAL INFORMATION

MAINTENANCE (RPV UPLOADED - GROUND LAUNCHER)

PREPARATION

ON LAUNCHER MAINTENANCE REQUIREMENTS

POST MAINTENANCE REQUIREMENTS AND CHECKOUT

EXTREME CLIMATIC AND ENVIRONMENTAL INFORMATION

SECTION VI POST RECOVERY

GENERAL

MID-AIR RETRIEVAL (MARS) RECOVERY

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UPLOAD, PRE/POSTFLIGHT AND DOWNLOAD MANUAL (Continued)

SECTION VI (Continued)

SURFACE RECOVERY

EQUIPMENT REQUIRED

TEST EQUIPMENT

TOOLS

CONSUMABLE MATERIALS

POST RECOVERY UNCOCK

PREPARATION

PROCEDURE

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3.3.13 Wiring Diagram Manual.

3.3.13.1 General Content. This manual shall contain cordage, wiring, installation, and schematic diagrams and associated data necessary for maintenance personnel to uderstand the make-up and function of each system and enable complete troubleshooting of the systems comprising the RPV.

3.3.13.2 Section I, Introduction and General Information. This section shall consist of at least the following primary paragraph headings; introduction, wire types, coax cables and waveguides and identification, and how to use the manual.

3.3.13.2.1 Introduction. The introduction shall briefly define construction and installation of wiring harness and cables.

3.3.13.2.2 Wire Types and Identification. Codes and symbols used on the diagrams shall be identified and explained. Codes and symbols shall be listed in tabular form with meanings and definition listed for each.

3.3.13.2.3 How to Use the Manual. Instructions for using the various diagrams contained in the Wiring Diagram manual shall be provided. Explanation of circuit tracing, continuity checks, voltage readings etc. shall be included. Identification of test points shall be explained. Diagram effectivity shall be identified. Use of engineering drawings as diagrams shall be included.

3.3.13.3 Section II, Diagrams. A complete list of diagrams shall precede the first diagram. Diagrams shall be arranged in the sequence most logical for ease of use during troubleshooting (i.e., cordage diagrams shall precede schematic and wiring diagrams, etc).

3.3.13.4 Manual Outline. The following is a sample outline of the Wiring Diagram manual. Paragraph headings included in the outline are mandatory. Additional paragraph headings may be added to increase usefulness of the manual during troubleshooting. Sample construction of diagrams are shown in figures 10 through 14.

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WIRING DIAGRAM MANUAL

NOTE

Front matter shall be arranged as required by paragraph 3.1.8 of this specification.

SECTION I INTRODUCTION AND GENERAL INFORMATION

GENERAL

INTRODUCTION

WIRING HARNESSES AND CABLES

COAX CABLES AND WAVEGUIDES

WIRING INSTALLATION

WIRE TYPES AND IDENTIFICATION

WIRE IDENTIFICATION CODE

CIRCUIT FUNCTION AND LETTER DESIGNATION

SYMBOLS CHART

HOW TO USE THE MANUAL

CORDAGE DIAGRAMS

SCHEMATIC DIAGRAMS

WIRING DIAGRAMS

INSTALLATION DIAGRAMS

TEST POINTS

SECTION II DIAGRAMS

GENERAL

LIST OF DIAGRAMS

DIAGRAMS

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3.3.14 Operational Systems Test Manual.

3.3.14.1 General Content. This manual shall contain all systems tests/check-out procedures normally required to prepare the RPV for flight (turn-around). Engine tests, systems tests, auxiliary equipment tests shall be included. Testing and checks that are only required incident to accomplishment of maintenance tasks shall not be included as this type of test is normally a part of the applicable maintenance procedures.

3.3.14.2 Section I, Introduction and General Information. This section shall consist of at least three primary paragraph headings; introduction, tabular format, test application.

3.3.14.2.1 Introduction. The introduction shall identify the tests included in the manual and provide a brief purpose of each test.

3.3.14.2.2 Tabular Format. An explanation of the tabular format (figure 3) shall be provided. In cases where computerized self-test systems are used, data flow diagrams (figure 4) may be used in lieu of or in conjunction with the tabular format. In all cases manual actions shall be provided in tabular format per figure 3.

3.3.14.2.3 Test Application. A complete description of the intent and application of each test included in the manual shall be provided. Use of partial tests following maintenance or during troubleshooting shall be explained. Actions required when malfunctions occur during testing shall be identified.

3.3.14.3 Section II, Engine Tests. This section shall provide all essential information to permit accomplishment of bare engine and installed engine tests. Installed tests while RPV is uploaded on launch platform shall be included. This section shall consist of at least three primary paragraph headings; equipment required, bare engine tests, and installed engine tests.

3.3.14.3.1 Equipment Required. All equipment and materials required to accomplish engine testing shall be identified as required by paragraph 3.2.1.2 of this specification.

3.3.14.3.2 Bare Engine Tests. Procedures for operational check of the bare engine following maintenance or upon receipt shall be provided.

3.3.14.3.3 Installed Engine Tests. Procedures for operational check of the engine installed in the RPV shall be provided. Operation with RPV uploaded on each authorized launch platform shall be included.

3.3.14.4 Section III, Systems Tests. This section shall provide procedures for accomplishing complete checkout of RPV operational systems. Test procedures shall be sequenced to provide the most efficient checkout of the complete system. Test procedures shall be constructed in such a manner to permit accomplishment of only that portion of any test necessary to verify proper operation of the system following maintenance or complete system checkout to include payload interface as required by the maintenance situation. This section shall consist of a primary paragraph heading equipment required and a primary paragraph heading for each major system test. The test setup shall be included at the beginning of the section for all tests. In cases of computerized self-test equipment the test setup shall always precede the initial test procedure.

3.3.14.4.1 Equipment Required. All equipment and materials required to accomplish systems checkout shall be identified as required by paragraph 3.2.1.2 of this specification.

3.3.14.4.2 Test Procedures. Dependent upon tests to be included and types of test equipment used, primary and subordinate paragraph headings shall be arranged

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to cover all systems/equipment in the most efficient and logical sequence.

3.3.14.5 Section IV, Auxiliary Equipment Tests. This section shall provide operational tests for auxiliary equipment/systems which are optional and normally not used for every mission/flight (i.e., beacons, scoring systems, various pods, etc). This section shall be arranged in the same manner and sequence as applied to section III, except test setup shall precede each test.

3.3.14.6 Section V, Post Test Requirements. This section shall provide necessary procedures to disconnect and store test equipment and verify final RPV preparation for upload. This section shall consist of at least two primary paragraph headings; test equipment disconnect, and RPV preparation.

3.3.14.6.1 Equipment Disconnect. Procedures for disconnecting and securing test equipment shall be provided. Information contained in other manuals shall be referenced and not duplicated.

3.3.14.6.2 RPV Preparation. Procedures for connecting wiring harnesses removed for test, installing access panels and doors, and final preparation of RPV for upload shall be provided. Requirements for actions such as parachute installation, servicing, weight and balance, etc., shall be stated and applicable manuals referenced. Procedures contained in other manuals shall not be duplicated.

3.3.14.7 Manual Outline. The following is a sample outline of the Operational Systems Test manual. Actual arrangement and paragraph headings shall be dependent upon the systems/equipment to be tested and the type of test equipment used (i.e., manual, automated). Paragraph headings in section I are mandatory. In all cases, equipment required headings are mandatory. Other paragraph headings shall be constructed to match systems/equipment applicable to the specific RPV covered by the manual.

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OPERATIONAL SYSTEMS TEST MANUAL

NOTE

Front matter shall be arranged as required by paragraph 3.1.8 of this specification.

SECTION I INTRODUCTION AND GENERAL INFORMATION

GENERAL

INTRODUCTION

ENGINE TESTS

SYSTEMS TESTS

AUXILIARY EQUIPMENT TESTS

TABULAR FORMAT

STEP NUMBER

PROCEDURE/ACTION

INDICATION/REACTION

PURPOSE/DIAGNOSIS

TEST APPLICATION

SYSTEMS CHECKOUT

MALFUNCTIONS DURING TESTS

CHECKOUT FOLLOWING MAINTENANCE

SECTION II ENGINE TESTS

GENERAL

EQUIPMENT REQUIRED

TEST EQUIPMENT

TOOLS

BARE ENGINE TEST

PREPARATION

Test Setup

Safety Precautions

Initial Switch Positions

TEST PROCEDURE

INSTALLED ENGINE TEST

PREPARATION

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OPERATIONAL SYSTEMS TEST MANUAL (Continued)

SECTION II (Continued)

Test Setup

Safety Precautions

Initial Switch Positions

TEST PROCEDURE

SECTION III SYSTEMS TESTS

GENERAL

EQUIPMENT REQUIRED

TEST EQUIPMENT

TOOLS

ELECTRICAL SYSTEM TEST

PREPARATION

Test Setup

Equipment Application

TEST PROCEDURE

RECOVERY SYSTEM TEST

PREPARATION

Test Setup

Equipment Application

TEST PROCEDURE

FLIGHT CONTROL SYSTEM TEST

PREPARATION

Test Setup

Equipment Application

TEST PROCEDURE

MICROWAVE COMMAND GUIDANCE SYSTEM (MCGS) TEST

PREPARATION

Test Setup

Equipment Application

TEST PROCEDURE

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OPERATIONAL SYSTEMS TEST MANUAL (Continued)

SECTION III (Continued)

ECM SYSTEM TEST

PREPARATION

Test Setup

Equipment Application

TEST PROCEDURE

CHAFF POD TEST

PREPARATION

Test Setup

Equipment Application

TEST PROCEDURE

SCORER TESTS

PREPARATION

Test Setup

Equipment Application

TEST PROCEDURE

TV CAMERA TESTS

PREPARATION

Test Setup

Equipment Application

TEST PROCEDURE

PAYLOAD (ARMAMENT SYSTEM) TESTS

PREPARATION

Test Setup

Equipment Application

TEST PROCEDURE

SECTION IV AUXILIARY EQUIPMENT TESTS

GENERAL

EQUIPMENT REQUIRED

TEST EQUIPMENT

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OPERATIONAL SYSTEMS TEST MANUAL (Continued)

SECTION III (Continued)

TOOLS

BEACON TEST

PREPARATION

Test Setup

Equipment Application

TEST PROCEDURE

POST RECOVERY LOCATOR BEACON TEST

PREPARATION

Test Setup

Equipment Application

TEST PROCEDURE

SECTION V POST TEST REQUIREMENTS

GENERAL

TEST EQUIPMENT DISCONNECT

DISCONNECT PROCEDURE

SECURING TEST EQUIPMENT

RPV PREPARATION

WIRE HARNESS RECONNECTIONS

ACCESS PANELS/DOORS INSTALLATION

PARACHUTE INSTALLATION

FUEL AND OIL REQUIREMENTS

BATTERIES

WEIGHT AND BALANCE

UPLOAD PREPARATION

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3.3.15 Payload and Reconnaissance System Manual.

3.3.15.1 General Content. This manual shall provide descriptions, removal, installation, maintenance, servicing, ballast installation and troubleshooting procedures for payload and reconnaissance systems for use by organizational (flightline) maintenance personnel. Systems/equipment normally covered by this manual include; scoring systems, chaff pods, electronic countermeasures and defensive systems, armament systems, electronic collection systems, etc.

3.3.15.2 Procurement Option. At the option of the procuring activity, systems contained in this manual may be included in the utility systems manual. This normally would apply only when payload systems are relatively simple and would not warrant procurement of a separate manual. If this option is elected, the payload and reconnaissance systems will be added to the utility systems manual as separate sections with the same arrangement and detail as required for a separate manual. (Refer to paragraph 3.3.10, 1.2.)

3.3.15.3 Section I, Introduction and Description. This section shall consist of at least four primary paragraph headings; introduction, leading particulars, physical description and functional description. Limitations and personnel/equipment safety shall be emphasized.

3.3.15.3.1 Introduction. The introduction shall identify the systems/equipment covered by the manual.

3.3.15.3.2 Leading Particulars. The leading particulars for each system/equipment shall be provided.

3.3.15.3.3 Physical Description. A complete physical description shall be provided for each system covered. The physical description shall include physical characteristics, dimensions, weights, locations, design features and mounting methods.

3.3.15.3.4 Functional Description. A complete functional description, supported by functional block diagrams, shall be provided for each system/equipment covered. Interface with related RPV systems shall be included.

3.3.15.4 Section II, Removal and Installation. This section shall consist of at least three primary paragraph headings; equipment required, removal, installation.

3.3.15.4.1 Equipment Required. All equipment and materials required to accomplish removal and installation of payload and reconnaissance systems/equipment shall be identified as required by paragraph 3.2.1.2 of this specification.

3.3.15.4.2 Removal. Procedures for removal of payload and reconnaissance systems/equipment shall be provided as required by paragraph 3.2.1.3 of this specification.

3.3.15.4.3 Installation. Procedures for installation of payload and reconnaissance systems/equipment shall be provided as required by paragraph 3.2.1.3 of this specification.

3.3.15.5 Section III, Maintenance and Servicing. This section shall consist of at least three primary paragraph headings; equipment required, maintenance, servicing.

3.3.15.5.1 Equipment Required. All equipment and materials required for maintenance and servicing of payload and reconnaissance systems shall be identified as required by paragraph 3.2.1.2 of this specification.

3.3.15.5.2 Maintenance. Maintenance procedures shall include inspection, cleaning, adjustments, lubrication, perational checkout and extreme climatic and environmental

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information as required by paragraph 3.2.1.4 of this specification.

3.3.15.5.3 Servicing. Servicing required incident to maintenance shall be provided as required by paragraph 3.2.1.5 of this specification.

3.3.15.6 Section IV, Troubleshooting. This section shall consist of at least two primary paragraph headings; inspection and systems analysis.

3.3.15.6.1 Inspection. Procedures for accomplishing visual inspection of components as a part of troubleshooting shall be provided as required by paragraph 3.2.1.6 of this specification.

3.3.15.6.2 Systems Anslysis. Troubleshooting procedures shall be tabulated as required by paragraph 3.2.1.6 of this specification.

3.3.15.7 Manual Outline. The following is a sample outline of the Payload and Reconnaissance System manual. Primary sideheads are mandatory. Subordinate sideheads under maintenance are mandatory. Other paragraph headings shall be dependent upon systems/equipment applicable to the specific RPV being covered by the manual.

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PAYLOAD AND RECONNAISSANCE SYSTEM MANUAL

NOTE

Front matter shall be arranged as required by paragraph 3.1.8 of this specification.

SECTION I INTRODUCTION AND DESCRIPTION

GENERAL

INTRODUCTION

SCORING SYSTEM

DEFENSIVE SYSTEM

BOMBING SYSTEM

LEADING PARTICULARS

PHYSICAL DESCRIPTION

SCORING SYSTEM

Scorer

Mounting

Material

DEFENSIVE SYSTEM

Electronic Countermeasures

Chaff Pods

BOMBING SYSTEM

Guided Missile

Launch Rack

Aerial Bomb

Bomb Rack

FUNCTIONAL DESCRIPTION

SCORING SYSTEM

DEFENSIVE SYSTEM

BOMBING SYSTEM

SECTION II REMOVAL AND INSTALLATION

GENERAL

EQUIPMENT REQUIRED

TEST EQUIPMENT

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PAYLOAD AND RECONNAISSANCE SYSTEM MANUAL (Continued)

SECTION II (Continued)

HANDLING EQUIPMENT

TOOLS

CONSUMABLE MATERIALS

REMOVAL

SCORER

SCORER MOUNT

ECM TRANSMITTER

ECM ANTENNAS

CHAFF PODS

MISSILE LAUNCH RACK

MISSILE

BOMB RACK

BOMBS

INSTALLATION

SCORER

SCORER MOUNT

ECM TRANSMITTERS

ECM ANTENNAS

CHAFF PODS

MISSILE LAUNCH RACK

MISSILE

BOMB RACK

BOMBS

SECTION III MAINTENANCE AND SERVICING

GENERAL

EQUIPMENT REQUIRED

TEST EQUIPMENT

HANDLING EQUIPMENT

TOOLS

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PAYLOAD AND RECONNAISSANCE SYSTEM MANUAL (Continued)

SECTION III (Continued)

CONSUMABLE MATERIALS

MAINTENANCE

INSPECTION

CLEANING

ADJUSTMENTS

LUBRICATION

OPERATIONAL CHECKOUT

EXTREME CLIMATIC AND ENVIRONMENTAL INFORMATION

SERVICING

SECTION IV TROUBLESHOOTING

GENERAL

INSPECTION

SYSTEM ANALYSIS

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4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection. Methods of inspection and approval shall be in accordance with MIL-M-38784.

5. PACKAGING

5.1 Packaging, packing and marking for shipment shall be in accordance with MIL-M-38784 and MIL-P-38790.

6. NOTES

6.1 Intended Use. The manuals to be prepared in accordance with this specification are to be used for organizational maintenance of remotely piloted vehicles and installed equipment.

6.2 Ordering Data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Using command will assist in determining content and complexity of coverage required.

c. Requirements for submittal of manuscript copy.

d. Requirements for review conferences including dates and locations.

e. Approved deviations to this specification and identify any additional requirements.

f. Whether the maintenance instructions which this specification covers shall be in several manuals, or in a single manual. (3.1.1 and 3.1.3)

g. If maintenance instructions for payload systems shall be provided in a separate manual or combined in the utility systems manual when multi-manual data is specified. (3.3.15.2 and 3.3.10.1.2)

h. Whether classified data shall be provided in separate manuals or included in the RPV General manual. (3.1.9 and 3.3.i.2)

6.3 Definitions. Terms used in or applicable to this specification are defined below.

6.3.1 Organizational (Flight Line) Maintenance. Organizational (flightline) maintenance is that maintenance authorized for, performed by, and the responsibility of using organizations on equipment while installed in a RPV. This maintenance consists of isolating malfunctions of a system to a component, accessory, assembly, subassembly, plug-in unit, etc; the removal and replacement of these items from thr RPV, performing any necessary alignments, adjustments or calibrations on installed equipment required as a result of such replacements, performing inspections and lubrication, and the tests necessary to determine if an installed equipment meets the minimum operational standard established for it. It does not include repair or checkout of components removed from the RPV.

6.3.2 Intermediate Level Maintenance. Intermediate level maintenance is that maintenance accomplished on a component, accessory, assembly, subassembly, plug-in unit etc after it is removed from the RPV. This maintenance is designed to return the item to serviceable condition. Intermediate level maintenance is normally accomplished in a shop area and includes repair, adjustment, bench check, etc.

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6.3.3 Depot Level Maintenance. Depot level maintenance is that maintenance that is beyond the capability of the organizational/field level activities. Depot level maintenance normally consists of overhaul, recondition, manufacture, extensive repair and modification, and technical assistance.

6.3.4 Validation. The process by which the contractor tests a technical order for technical accuracy. Validation may be accomplished at the contractors facility, or if directed by the procurement activity at an operational site. Validation entails the actual performance of operating and maintenance procedures by the contractor.

6.3.5 Verification. The process by which a technical order is tested and proven by Air Force personnel under Air Force jurisdiction for adequacy of operation and maintenance of equipment procured for operational units. It also certifies that selected technical orders are compatible with the hardware. Verification consists of actual performance of operating and maintenance procedures by using command personnel.

6.3.6 Systems. A system is defined as any combination of complete operating equipments, components, accessories, or detail parts interconnected or interrelated in such a manner so as to perform a specific operational function or functions.

6.3.7 Equipment. An equipment shall consist of one or more components capable of performing specified functions.

6.3.8 Module. A module is defnined as a subassembly which, in the area of electronic systems, may be removed and replaced without use of soldering equipment or special tools, a module may be encapsulated.

6.3.9 Component. Component is defined as an article which is normally a combination of detail parts, subassemblies, and assemblies, and is a self-contained element of a complete operating equipment and performs a function necessary to the operation of that equipment.

Examples: A wing is a component of the RPV, while the wing tip is a component of the wing. Other examples are indicator unit, power unit, receiver, transmitter tuning units, rotating antenna, modulator unit, amplifier unit, etc.

6.3.10 Assembly. An assembly is defined as an article which consists of detail parts and subassemblies, or a combination thereof, and as such is an element of a component, and performs functions necessary to the operation of rf, if video, etc., gyro assembly, oscillator assembly (rf or af) pulsing networks, etc. The following type items are assemblies only when an integral part of a component. Examples: filters, amplifier modulators, power supplies, junction boxes, etc.

6.3.11 Subassembly. A subassembly is defined as an article which is an element of an assembly, replaceable as a whole but also having some parts individually replaceable, consisting of a combination of detail parts having a common mounting or mounted on each other. Examples: wafer switch, variable capacitor, coil and associated tuning assembly (such as an i-f transformer with tuning slug and mechanism), terminal board with parts assembled thereon, etc. The following type items are subassemblies only when part of an assembly. Examples: amplifier section, electrical instrument, transformer, variometer, vibrator, voltage regulator, dynamotor machine, etc.

6.3.12 Accessory. An accessory is defined as any article not normally procured as part of an equipment but which is ordinarily requiree to permit assembly,

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operation, or maintenance of an equipment or system. An accessory may be common to several equipments. Examples: emergency power units, headset, microphone, hand key, extension cord and cable, operator's chair, plotting board, antenna reel, etc.

6.3.13 Nomenclature. The nomenclature of equipment or components is the official name as assigned by the government. The following is an example of the correct form of nomenclature of a radio receiver which has been assigned Air Force-Navy nomenclature: Receiver, Radio R-484/AFR-14. The nomenclature is a manual shall always agree with the nomenclature used on the applicable engineering drawing. Once properly identified with a short or common nomenclature, the short term may be used.

6.3.14 Use of Words Shall and Will. The use of the words shall and will in this specification indicates a mandatory requirement.

6.3.15 Use of Word Should. The use of the word should in this specification indicates a non-mandatory desired or preferred method.

6.3.16 Use of Word May. The use of the word may in this specification indicates an acceptable or suggested method.

CUSTODIAN:

Air Force - 11

REVIEW ACTIVITIES:

Air Force - 01, 10, 99

PREPARING ACTIVITY:

Air Force - 16

PROJECT NO.

TMSS F432

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CONFIGURATION CHART

TO ISSUE	AFTO FORM 22, OS/IOS	PRODUCTION EFFECTIVITY	ECP/TCTO

Figure 1. Configuration Page

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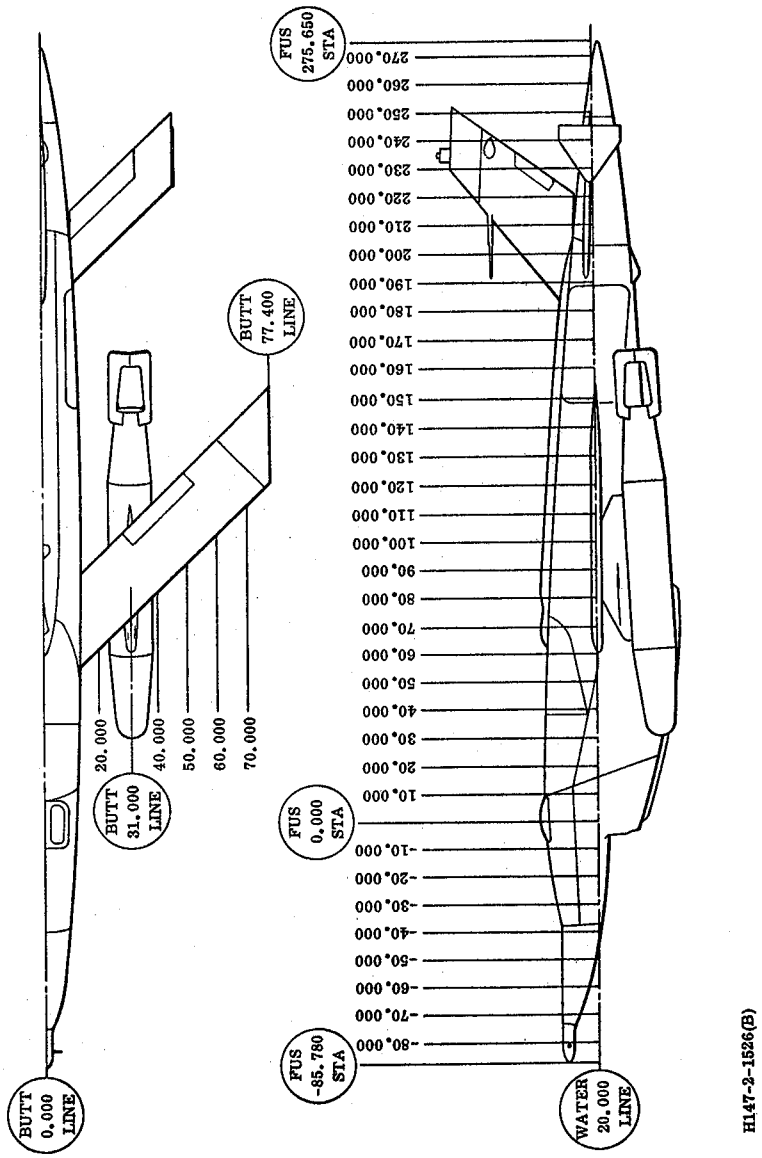


Figure 2. Station Locator Diagram

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STEP NO.	PROCEDURE/ACTION	INDICATION/REACTION	PURPOSE/DIAGNOSIS
	SECTION A. PYLON BATTERY POWER TEST		
	NOTE When DVM is used, set DVM controls to accept specific test. This instruction will not be repeated for every step.		
1	Rotate DVM switch (switching panel) to DIR position	a. PYLON BATT light (DCP) on approximately 10 seconds	Function of switch action only
		b. DVM indicates 24.5 to 29.0 Vdc	To clear DVM
	SECTION B. PRIMARY POWER DISTRIBUTION TESTS		
1	Close AC and DC circuit breakers on ATP	No indication	To ensure actuators are in a braked condition
2	Rotate TPS switch to position 26	CONTROL light on	To prevent crossover during tests

Figure 3. Sample Procedure Format

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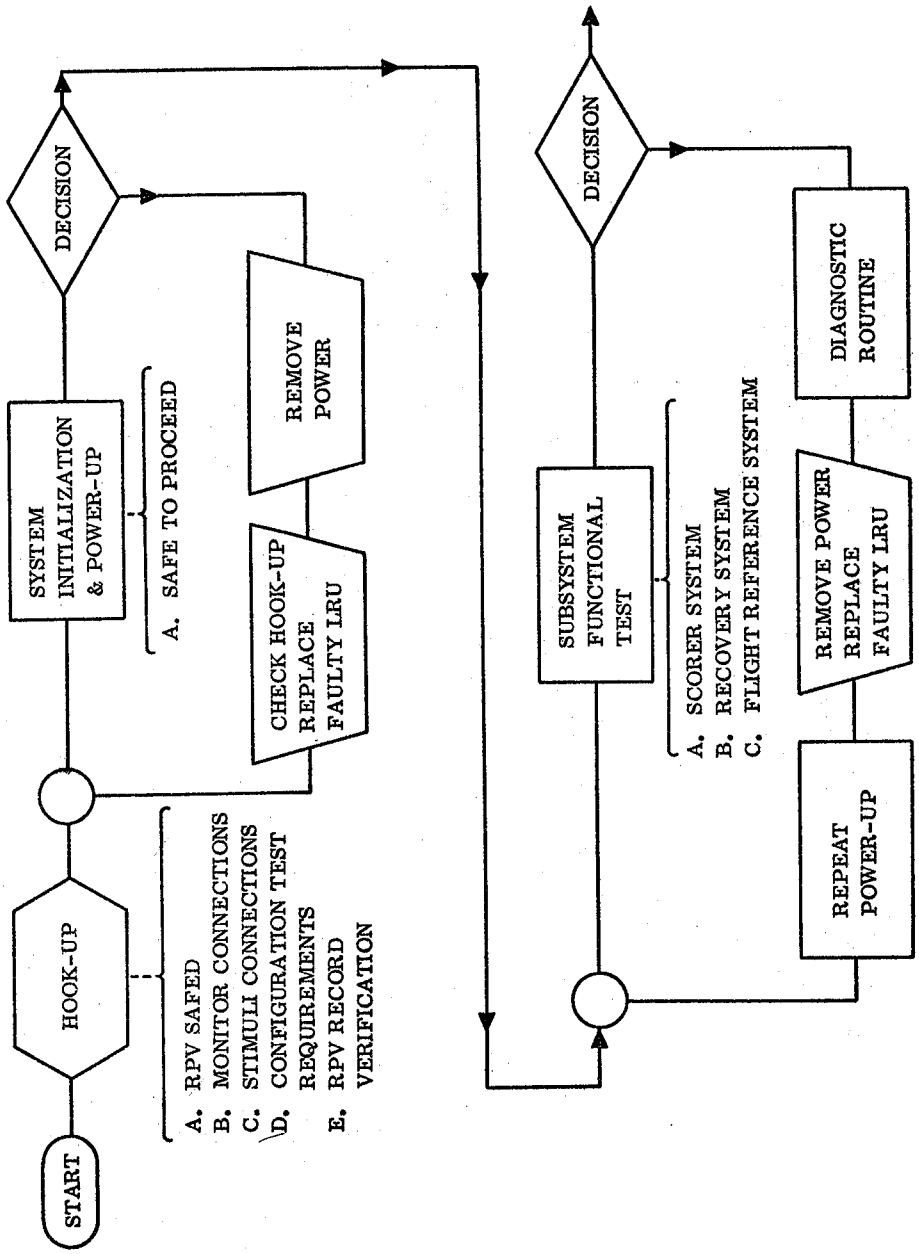


Figure 4. Automatic Systems Test Flow Chart

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During accomplishment of removal and installation instructions contained in this section, standard maintenance and safety practices must be observed. The following items are applicable to the procedures contained in this section.

1. Electrical power shall be removed before starting any component removal or installation action.

2. Protective dust covers or caps shall be installed on all electrical connectors or open lines when they are disconnected. Covers and caps will be removed and connectors and open lines inspected for dirt or damage prior to being connected. Applicable safetywire or safety nuts shall be installed.

3. Specific torque values are identified in the procedures. When no torque value is indicated for nuts, screws, bolts, or clamps, the standard torque is applicable for the item being installed. Refer to T.O. 1-1A-8 and T.O. 32B14-3-1-101 to determine standard torque values and torque procedures.

4. Each replacement component shall be inspected and determined to be serviceable before installation. The part number on a replacement component shall be compared with the part number on the removal component to ensure it is a correct replacement for the item removed.

5. Documentation of component replacement shall be accomplished as required by T.O. 00-20- series technical orders.

6. Standard safety procedures required by AFR 127-101 shall be observed during each replacement action.

7. When a system is disturbed by maintenance, such a component removed or disconnected, which could result in equipment damage or personnel injury by application of power, the following action shall be accomplished.

a. Annotate AFTO Form 781 to identify the condition.

b. Attach Danger Tag (AF Form 1492 or equivalent) to main umbilical or other area clearly visible to maintenance personnel.

8. Conduct an operational test upon replacement or repair of any components.

Figure 5. Sample Introductory Paragraph for Removal/Installation Procedures

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EQUIPMENT	NO. OF UNITS	AMPS PER UNIT	OPERATING TIME	GENERATOR POWER 28V																BATTERY POWER 28V																	
				OPERATING CONDITIONS																																	
				START AND WARM-UP				LAUNCH AND CLIMB				CRUISE				GLIDE				DRAG CHUTE				MAIN CHUTE				RECOVERY									
				AVERAGE AMPS				AVERAGE AMPS				AVERAGE AMPS				AVERAGE AMPS				AVERAGE AMPS				AVERAGE AMPS				AVERAGE AMPS									
				MAX AMPS	0.5 MIN	2.0 MIN	4.0 MIN	MAX AMPS	0.5 MIN	2.0 MIN	25.0 MIN	MAX AMPS	0.5 MIN	40.0 MIN	15.0 MIN	MAX AMPS	0.5 MIN	2.0 MIN	5.0 MIN	MAX AMPS	0.01 MIN	0.3 MIN	3.0 MIN	MAX AMPS	0.7 MIN	2.0 MIN	15.0 MIN	MAX AMPS	MARS CONT	0.07 MIN	LAND CONT	SEA CONT						
CONTROL SURFACE:																																				
FLIGHT CONTROL BOX	1	3.00	CONT	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	—	—	—	—	—	—	—	—	—	—	—	—						
AILERON ACTUATOR	1	2.50	CONT	2.50	2.50	2.50	2.50	2.50	2.50	2.50	2.50	2.50	2.50	2.50	2.50	2.50	2.50	2.50	2.50	—	—	—	—	—	—	—	—	—	—	—	—						
RUDDER ACTUATOR	1	2.50	CONT	2.50	2.50	2.50	2.50	2.50	2.50	2.50	2.50	2.50	2.50	2.50	2.50	2.50	2.50	2.50	2.50	—	—	—	—	—	—	—	—	—	—	—	—						
RUDDER TRIM	1	5.00	MOM	5.00	—	—	—	5.00	0.10	0.10	0.10	5.00	0.05	0.05	0.05	5.00	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—						
FLIGHT INSTRUMENTS:																																				
MCGS	1	2.50	CONT	2.50	2.50	2.50	2.50	2.50	2.50	2.50	—	2.50	—	—	2.50	2.50	2.50	2.50	2.50	2.50	2.50	2.50	2.50	2.50	2.50	2.50	2.50	2.50	2.50	2.50	—					
PROGRAMMER	1	1.00	CONT	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—						
INVERTER, 750VA, 3Ø	1	37.60	CONT	37.60	37.60	37.60	37.60	37.60	37.60	37.60	37.60	37.60	37.60	37.60	37.60	25.90	25.90	25.90	25.90	—	—	—	—	—	—	—	—	—	—	—	—						
INVERTER, 750VA, 1Ø	1	26.50	CONT	26.50	26.50	26.50	26.50	26.50	26.50	26.50	26.50	26.50	26.50	26.50	26.50	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—						
CONVERTER RPM	1	0.04	CONT	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	—	—	—	—	—	—	—	—						
SIGNAL CONDITIONER	1	0.10	CONT	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.10	—	—	—	—	—	—	—	—						
CONTROL-FUEL, OIL & PRESS:																																				
THROTTLE ACTUATOR	1	0.75	MOM	0.75	0.25	0.25	0.25	0.75	0.3	0.13	0.13	0.75	0.07	0.07	0.07	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—						
FUEL BOOST PUMP	1	16.00	CONT	16.00	10.00	12.00	12.00	16.00	16.00	16.00	16.00	16.00	14.00	14.00	14.00	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—						
FUEL SHUTOFF VALVE	1	1.00	CONT	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—						
FUEL VENT VALVE	1	0.54	CONT	—	—	—	—	0.54	—	—	0.54	0.54	—	0.54	0.54	0.54	0.54	0.54	—	—	—	—	—	—	—	—	—	—	—	—	—						
OIL VENT VALVE	1	1.10	CONT	—	—	—	—	—	—	—	—	1.10	1.10	1.10	1.10	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—						
BLEED AIR VALVE	1	0.95	CONT	—	—	—	—	—	—	—	—	0.95	0.95	0.95	0.95	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—						
EQUIP COMP VENT VALVE	1	1.00	CONT	—	—	—	—	—	—	—	—	1.00	1.00	1.00	1.00	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—						
ANTI BLEED VALVE	1	0.95	CONT	—	—	—	—	—	—	—	—	0.95	0.95	0.95	0.95	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—						
FUEL EXPULSION VALVE	1	1.70	MOM	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	1.70	1.70	—	—	1.70	—	1.70	—	—					
CONTROL POWER:																																				
POWER CONTROL BOX	1	1.00	CONT	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	—	—	—	—	—	—	—	—	—	—	—	—	0.40	—	—	0.40	0.40					
SPL DEVICES CONTROL BOX	1	1.20	CONT	0.40	0.40	0.40	0.40	0.40	0.40	0.40	0.40	0.40	0.40	0.40	0.40	0.40	0.40	0.40	0.40	0.40	0.40	0.40	0.40	0.80	0.80	0.80	0.80	0.80	0.80	0.80	—	—				
SQUIB CONTROL BOX	1	0.20	CONT	—	—	—	—	0.20	0.20	0.20	0.20	0.20	0.20	0.20	0.20	0.20	0.20	0.20	0.20	0.40	0.40	0.40	0.40	0.60	0.60	0.60	0.60	0.70	0.70	0.70	—	—				
RECOVERY:																																				
CHUTE CONTROL TIMER	2	0.13	CONT	0.26	0.26	0.26	0.26	0.26	0.26	0.26	0.26	0.26	0.26	0.26	0.26	0.26	0.26	0.26	0.26	0.26	0.26	0.26	0.26	0.26	0.26	0.26	0.26	0.26	0.26	0.26	—	—				
RECOVERY RELAY BOX	1	2.14	CONT	0.40	0.40	0.40	0.23	1.35	1.35	1.35	1.28	1.50	1.50	1.50	1.50	1.58	1.52	1.52	1.58	1.98	1.65	1.98	1.98	2.09	2.09	2.09	2.09	2.14	2.14	2.14	—	—				
BEACON RELAY	1	0.04	CONT	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	—	—				
DRAG PRIMER	4	9.00	MOM	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	*36.00	*36.00	—	—	—	—	—	—	—	—	—	—	—				
MAIN PRIMER	6	9.00	MOM	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	*54.00	*54.00	—	—	*9.00	—	*9.00	—	—				
SWIVEL PRIMER	1	9.00	MOM	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	*9.00	*9.00	—	—	—				
STAB CHUTE GUILLOTINE	1	9.00	MOM	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	*9.00	*9.00	—	—	—				
AUTO RECOVERY BOX	1	0.23	CONT	—	—	—	—	0.11	0.11	0.11	0.05	0.23	0.23	0.17	0.23	0.11	0.11	0.11	0.05	0.23	0.23	0.23	0.16	0.12	0.12	0.12	0.12	0.12	0.12	0.12	—	—				
HEATERS:																																				
BATTERY	1	2.50	50%	2.50	2.50	2.50	2.50	2.50	1.25	1.25	1.25	2.50	1.25	1.25	1.25	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—						
PITOT TUBE	1	7-40	CONT	40.00	25.00	19.00	16.00	16.00	13.00	13.00	13.00	16.00	10.00	10.00	10.00	16.00	14.00	14.00	12.00	14.00	10.00	10.00	10.00	—	—	—	—	—	—	—	—						
ANGLE OF ATTACK	1	5.00	CONT	5.00	5.00	5.00	5.00	5.00	5.00	5.00	5.00	5.00	5.00	5.00	5.00	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—						
SCORER FAN & HTR SYS	1	19.18	50%	19.18	19.18	19.18	19.18	19.18	19.18	18.18	9.59	18.18	18.18	9.59	9.59	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—						
AIR TEMP HTR	1	5-20	CONT	20.00	12.00	7.00	6.00	10.00	8.00	9.30	9.30	9.30	9.30	9.30	9.30	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—						
AMMONIA TANK	1	12.50	50%	12.50	12.50	12.50	12.50	12.50	12.50	12.50	12.50	12.50	12.50	6.25	6.25	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—						
ELECTRONICS:																																				
** SCORING SYS KS-120A	1	1.43	CONT	1.43	—	—	—	1.43	1.43	1.43	1.43	1.43	1.43	1.43	1.43	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—						
** SCORING SYS F415Y		2.06	CONT	2.06	—	—	—	2.06	2.06	2.06	2.06	2.06	2.06	2.06	2.06	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—						
DOPPLER SYSTEM	1	0.50	CONT	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	—	—	—	—	—	—	—	—	—	—	—	—						
L BAND BEACON	1	2.4	CONT	—	—	—	—	—	—	—	—	2.4	2.4	1.2	2.4	—	—	—	—	2.4	2.4	2.4	2.4	2.4	2.4	2.4	2.4	2.4	2.4	24	24						
COOLING:																																				
FWD COOLER	1	7.0	50%	7.00	7.00	7.00	7.00	7.00	7.00	7.00	3.50	7.00	7.00	3.50	3.50	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—						
AFT COOLER	1	7.0	CONT	7.00	7.00	7.00	7.00	7.00	7.00	7.00	7.00	7.00	7.00	7.00	7.00	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—						
TOTAL:				215.66	179.73	170.73	166.56	183.46	172.16	172.45	167.27	189.43	171.51	161.45	165.21	61.13	54.07	54.07	51.53	22.38	18.02	18.95	18.28	10.51	10.51	8.81	8.81	11.06	8.96	10.66	2.80	2.80				

* NOT INCLUDED IN TOTAL. DURATION OF LOAD TOO SHORT TO BE SIGNIFICANT.

** NOT INCLUDED IN TOTAL.

Figure 6. DC Electrical Load Analysis Chart

H147-2-1512(B)

107

MIL-M-83493(USAF)

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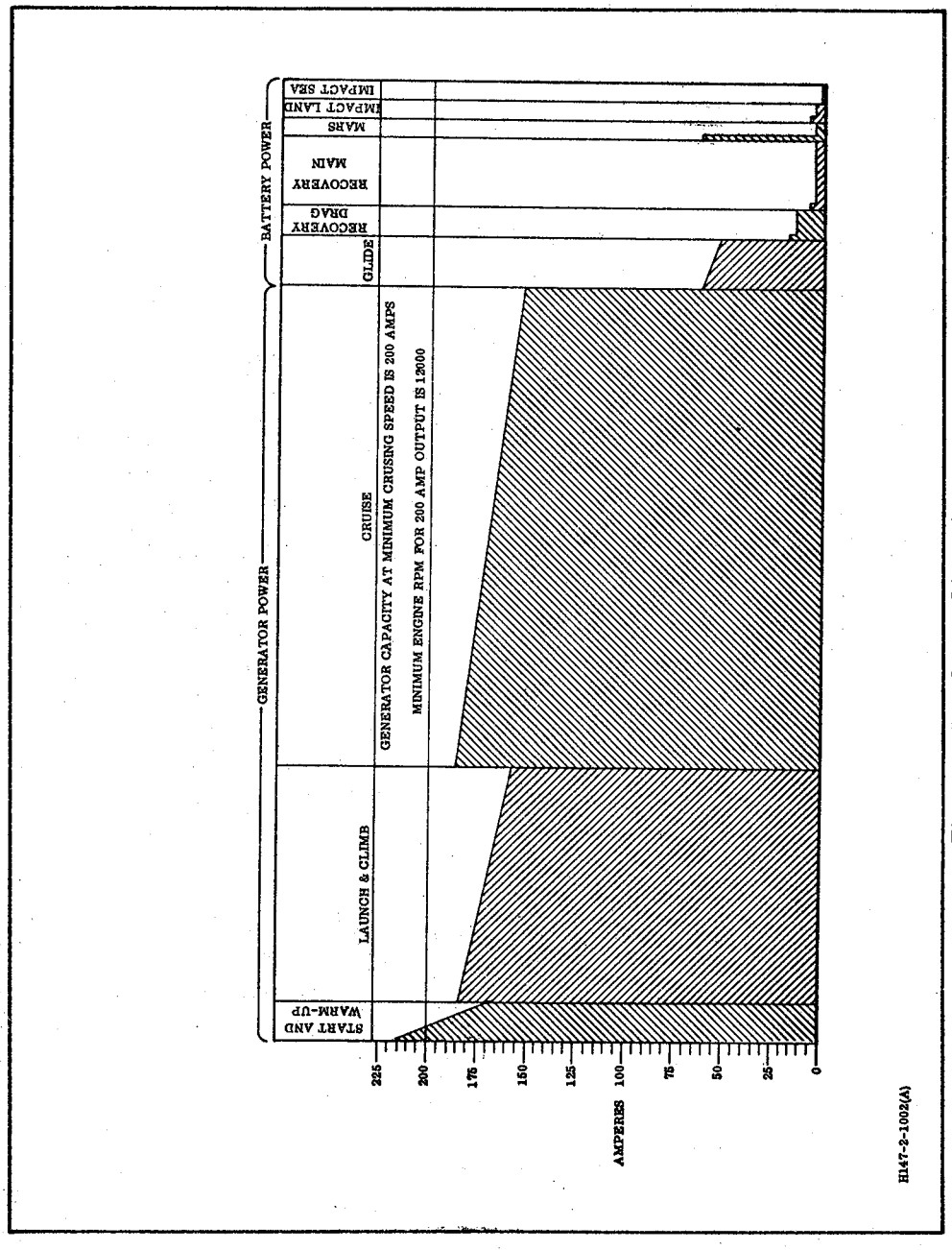
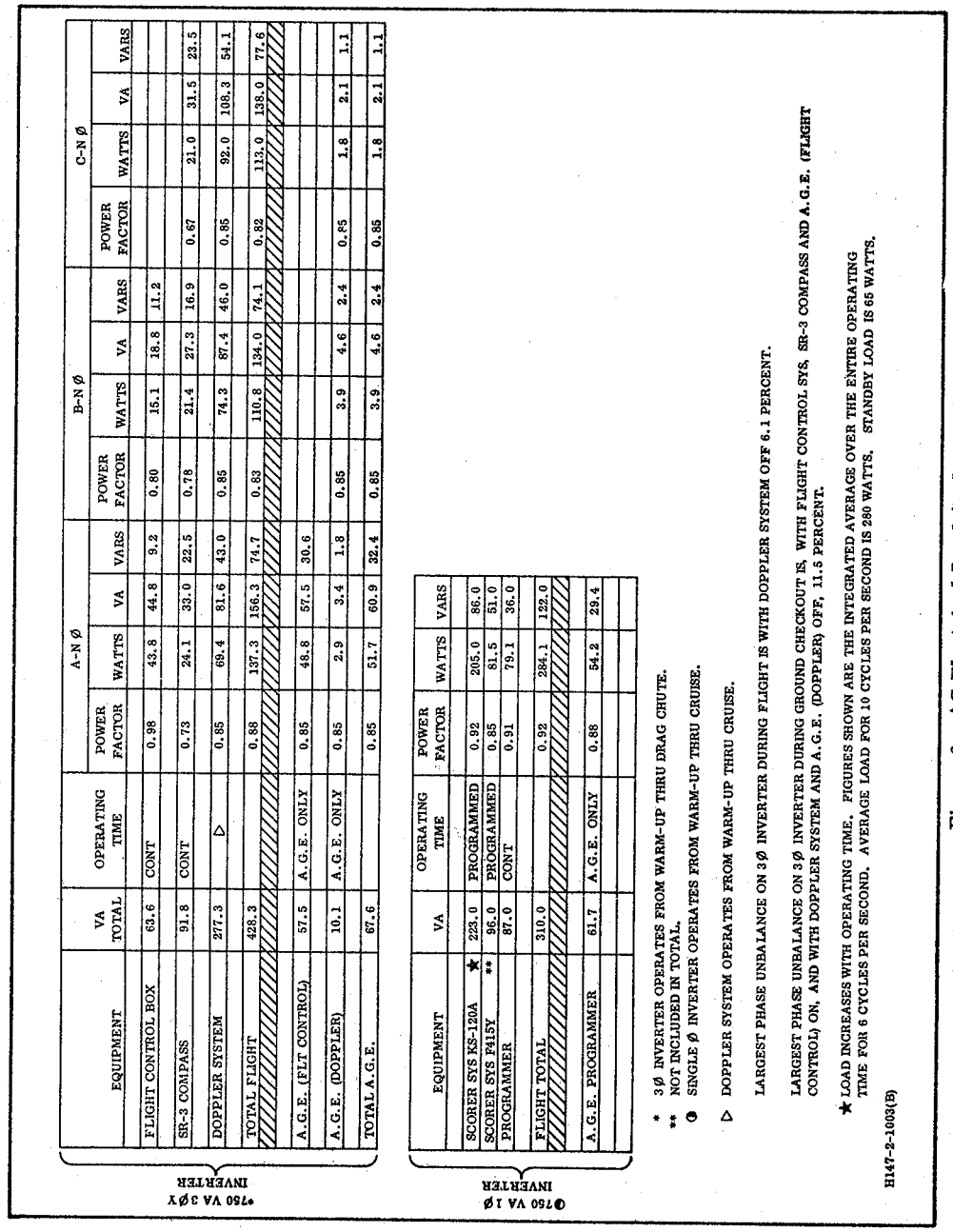


Figure 7. DC Electrical Load Analysis Graph

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* 3Ø INVERTER OPERATES FROM WARM-UP THRU DRAG CHUTE.

** NOT INCLUDED IN TOTAL.

● SINGLE Ø INVERTER OPERATES FROM WARM-UP THRU CRUISE.

△ DOPPLER SYSTEM OPERATES FROM WARM-UP THRU CRUISE.

LARGEST PHASE UNBALANCE ON 3Ø INVERTER DURING FLIGHT IS WITH DOPPLER SYSTEM OFF 6.1 PERCENT.

LARGEST PHASE UNBALANCE ON 3Ø INVERTER DURING GROUND CHECKOUT IS, WITH FLIGHT CONTROL SYS, SR-3 COMPASS AND A.G.E. (FLIGHT CONTROL) ON, AND WITH DOPPLER SYSTEM AND A.G.E. (DOPPLER) OFF, 11.5 PERCENT.

★ LOAD INCREASES WITH OPERATING TIME. FIGURES SHOWN ARE THE INTEGRATED AVERAGE OVER THE ENTIRE OPERATING TIME FOR 6 CYCLES PER SECOND. AVERAGE LOAD FOR 10 CYCLES PER SECOND IS 280 WATTS. STANDBY LOAD IS 65 WATTS.

H147-2-1003(B)

Figure 8. AC Electrical Load Analysis Chart

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STEP NO.	SYMPTOM	PROBABLE CAUSE	CORRECTIVE ACTION

NOTE

Troubleshooting procedures should coincide with procedures in Operational Systems Test manual. It is not necessary to relate each troubleshooting step to a corresponding step in the test procedure. Relationship to applicable test is sufficient.

Figure 9. Instructions for Systems Analysis (Troubleshooting)

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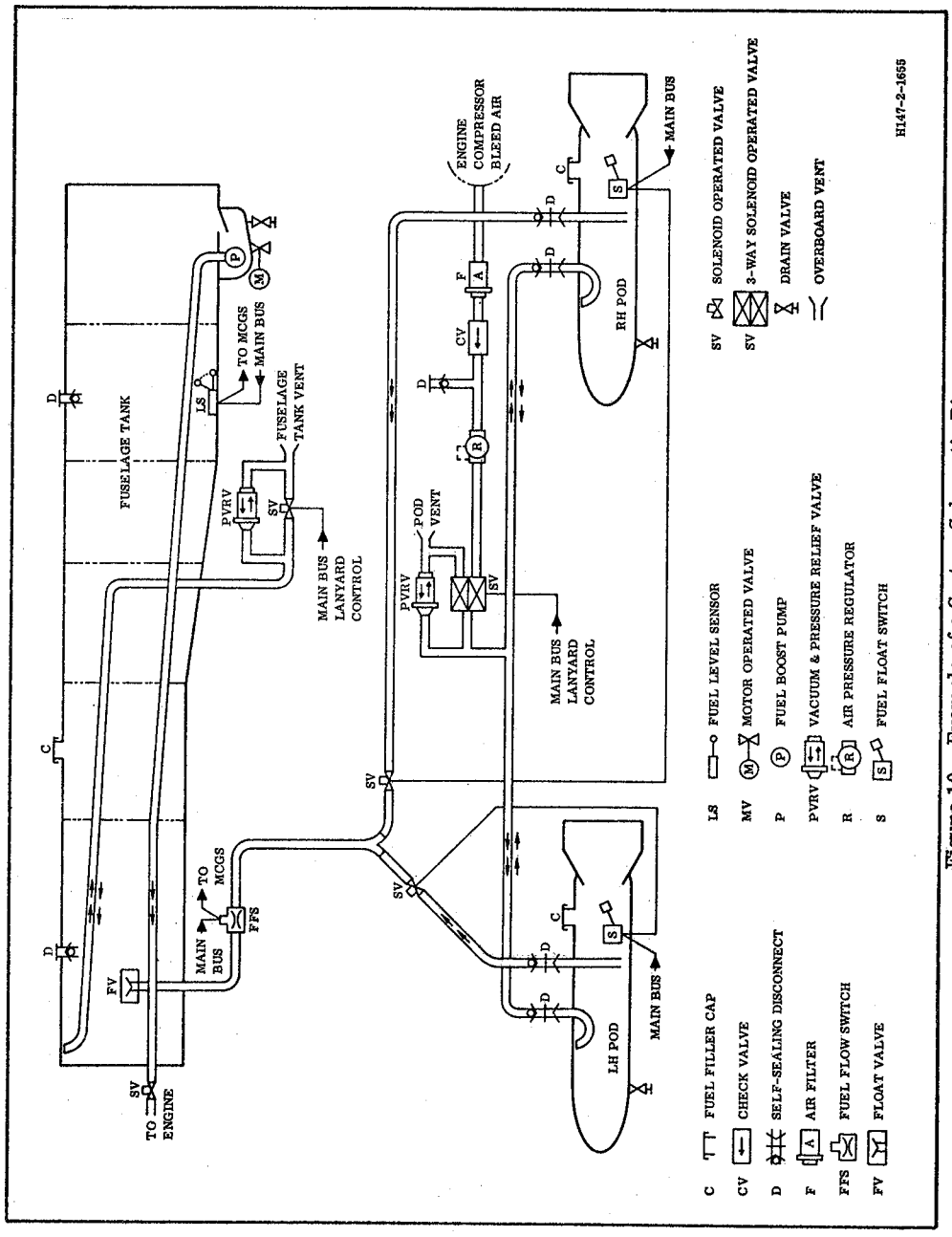


Figure 10. Example of a System Schematic Diagram

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The digital copy of MIL-M-83493, dated 15 Oct 1977 does not include foldout pages 113 thru 120. Please call the ASSIST HELP DESK at 215-697-6396 for copies of these pages.

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Data contained in this manual are applicable to AQM-34 RPV procured under Air Force contracts F33657-71-C-0462 and F33657-73-C-0558. Data peculiar to AQM-34R RPV in either category are identified by code symbols [A] or [B] respectively which appear adjacent to paragraph, procedural step, test response, figure, table, index or part number, note, caution or warning. Data not identified by code symbols are applicable to all configurations AQM-34R RPV. Additional RPV/code symbol explanation follows:

Code	Model	Air Force Serial Number
[A]	AQM-34R	AF67-21593, AF69-6114, AF69-6123 thru AF69-6130
[B]	AQM-34R	AF67-21575, AF67-21580, AF67-21581, AF69-6106, AF69-6131 thru AF69-6136, AF67-21589

NOTE

Subsequent configurations shall be identified by using additional code letters [C] [D] [E] etc.

Figure 15. Sample Configuration Coding Procedure

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